

Serial Communication Interface (SCI) Module

This chapter contains the description of the serial communication interface (SCI) module.

Topic	Page
29.1 Introduction	1609
29.2 SCI Communication Formats.....	1611
29.3 SCI Interrupts	1616
29.4 SCI DMA Interface.....	1619
29.5 SCI Configurations.....	1620
29.6 SCI Low-Power Mode	1621
29.7 SCI Control Registers	1623
29.8 GPIO Functionality	1653

29.1 Introduction

The SCI module is a universal asynchronous receiver-transmitter that implements the standard nonreturn to zero format. The SCI can be used to communicate, for example, through an RS-232 port or over a K-line.

29.1.1 SCI Features

The following are the features of the SCI module:

- Standard universal asynchronous receiver-transmitter (UART) communication
- Supports full- or half-duplex operation
- Standard nonreturn to zero (NRZ) format
- Double-buffered receive and transmit functions
- Supports two individually enabled interrupt lines: level 0 and level 1
- Configurable frame format of 3 to 13 bits per character based on the following:
 - Data word length programmable from one to eight bits
 - Additional address bit in address-bit mode
 - Parity programmable for zero or one parity bit, odd or even parity
 - Stop programmable for one or two stop bits
- Asynchronous communication mode with no CLK pin
- Two multiprocessor communication formats allow communication between more than two devices
- Sleep mode is available to free CPU resources during multiprocessor communication and then wake up to receive an incoming message
- The 24-bit programmable baud rate supports 2^{24} different baud rates provide high accuracy baud rate selection
- Capability to use Direct Memory Access (DMA) for transmit and receive data
- Four error flags and Five status flags provide detailed information regarding SCI events
- Two external pins: SCIRX and SCITX

NOTE: SCI module does not support UART Hardware Flow Control. This feature can be implemented in Software using a General Purpose I/O pin.

29.1.2 Block Diagram

Three Major components of the SCI Module are:

- Transmitter
- Baud Clock Generator
- Receiver

Transmitter (TX) contains two major registers to perform double buffering:

- The transmitter data buffer register (SCITD) contains data loaded by the CPU to be transferred to the shift register for transmission.
- The transmitter shift register (SCITXSHF) loads data from the data buffer (SCITD) and shifts data onto the SCITX pin, one bit at a time.

Baud Clock Generator

- A programmable baud generator produces a baud clock scaled from VCLK.

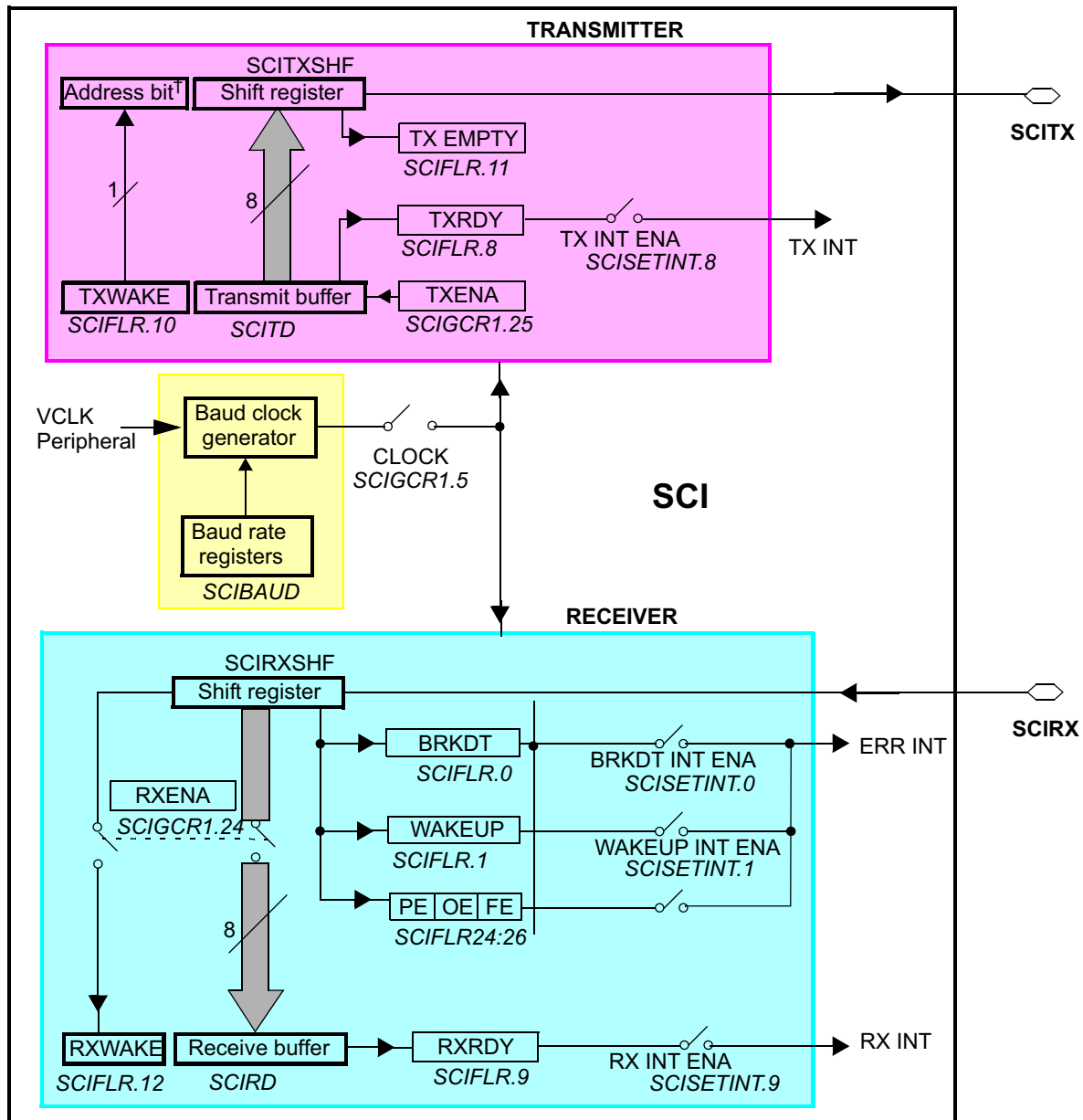
Receiver (RX) contains two major registers to perform double buffering:

- The receiver shift register (SCIRXSHF) shifts data in from the SCIRX pin one bit at a time and transfers completed data into the receive data buffer.
- The receiver data buffer register (SCIRD) contains received data transferred from the receiver shift register

The SCI receiver and transmitter are double-buffered, and each has its own separate enable and interrupt bits. The receiver and transmitter may each be operated independently or simultaneously in full duplex mode.

To ensure data integrity, the SCI checks the data it receives for breaks, parity, overrun, and framing errors. The bit rate (baud) is programmable to over 16 million different rates through a 24-bit baud-select register. Figure 29-1 shows the detailed SCI block diagram.

Figure 29-1. Detailed SCI Block Diagram



29.2 SCI Communication Formats

The SCI module can be configured to meet the requirements of many applications. Because communication formats vary depending on the specific application, many attributes of the SCI are user configurable. The list below describes these configuration options:

- SCI Frame format
- SCI Timing modes
- SCI Baud rate
- SCI Multiprocessor modes

29.2.1 SCI Frame Formats

The SCI uses a programmable frame format. All frames consist of the following:

- One start bit
- One to eight data bits
- Zero or one address bit
- Zero or one parity bit
- One or two stop bits

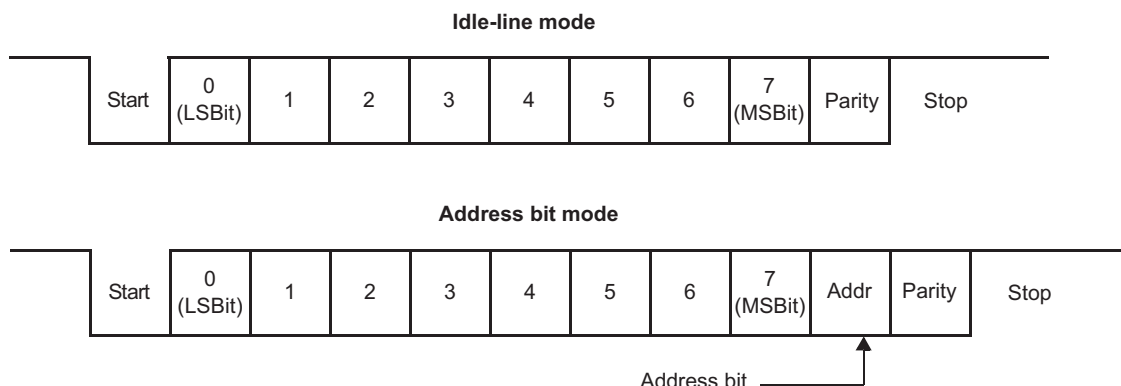
The frame format for both the transmitter and receiver is programmable through the bits in the SCIGCR1 register. Both receive and transmit data is in nonreturn to zero (NRZ) format, which means that the transmit and receive lines are at logic high when idle. Each frame transmission begins with a start bit, in which the transmitter pulls the SCI line low (logic low). Following the start bit, the frame data is sent and received least significant bit first (LSB).

An address bit is present in each frame if the SCI is configured to be in address-bit mode but is not present in any frame if the SCI is configured for idle-line mode. The format of frames with and without the address bit is illustrated in [Figure 29-2](#).

A parity bit is present in every frame when the PARITY ENA bit is set. The value of the parity bit depends on the number of one bits in the frame and whether odd or even parity has been selected via the PARITY ENA bit. Both examples in [Figure 29-2](#) have parity enabled.

All frames include one stop bit, which is always a high level. This high level at the end of each frame is used to indicate the end of a frame to ensure synchronization between communicating devices. Two stop bits are transmitted if the STOP bit in SCIGCR1 register is set. The examples shown in [Figure 29-2](#) use one stop bit per frame.

Figure 29-2. Typical SCI Data Frame Formats



29.2.2 SCI Timing Mode

The SCI can be configured to use asynchronous or isosynchronous timing using TIMING MODE bit in SCIGCR1 register.

29.2.2.1 Asynchronous Timing Mode

The asynchronous timing mode uses only the receive and transmit data lines to interface with devices using the standard universal asynchronous receiver-transmitter (UART) protocol.

In the asynchronous timing mode, each bit in a frame has a duration of 16 SCI baud clock periods. Each bit therefore consists of 16 samples (one for each clock period). When the SCI is using asynchronous mode, the baud rates of all communicating devices must match as closely as possible. Receive errors result from devices communicating at different baud rates.

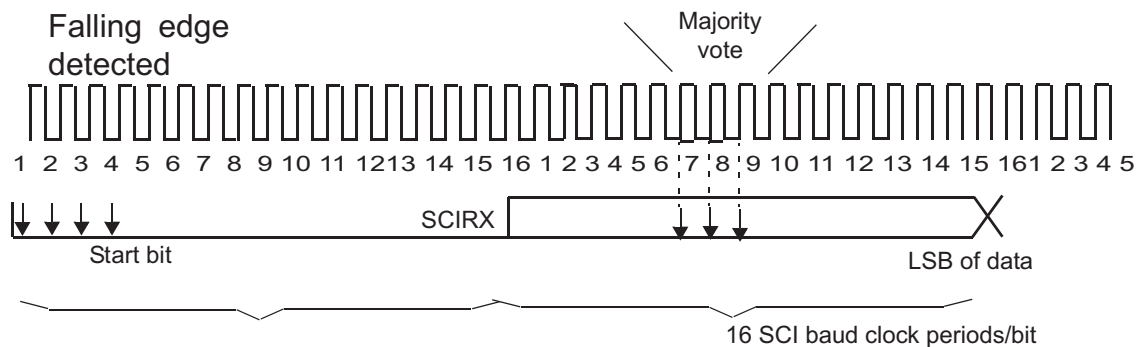
With the receiver in the asynchronous timing mode, the SCI detects a valid start bit if the first four samples after a falling edge on the SCIRX pin are of logic level 0. As soon as a falling edge is detected on SCIRX, the SCI assumes that a frame is being received and synchronizes itself to the bus.

To prevent interpreting noise as Start bit SCI expects SCIRX line to be low for at least four contiguous SCI baud clock periods to detect a valid start bit. The bus is considered idle if this condition is not met. When a valid start bit is detected, the SCI determines the value of each bit by sampling the SCIRX line value during the seventh, eighth, and ninth SCI baud clock periods. A majority vote of these three samples is used to determine the value stored in the SCI receiver shift register. By sampling in the middle of the bit, the SCI reduces errors caused by propagation delays and rise and fall times and data line noises.

Figure 29-3 illustrates how the receiver samples a start bit and a data bit in asynchronous timing mode.

The transmitter transmits each bit for a duration of 16 SCI baud clock periods. During the first clock period for a bit, the transmitter shifts the value of that bit onto the SCITX pin. The transmitter then holds the current bit value on SCITX for 16 SCI baud clock periods.

Figure 29-3. Asynchronous Communication Bit Timing



29.2.2.2 Isosynchronous Timing Mode

In isosynchronous timing mode, each bit in a frame has a duration of exactly 1 baud clock period and therefore consists of a single sample. With this timing configuration, the transmitter and receiver are required to make use of the SCICLK pin to synchronize communication with other SCI. **This mode is not fully supported on this device because SCICLK pin is not available.**

29.2.3 SCI Baud Rate

The SCI has an internally generated serial clock determined by the peripheral VCLK and the prescalers BAUD. The SCI uses the 24-bit integer prescaler BAUD value of the BRS register to select the required baud rates.

In asynchronous timing mode, the SCI generates a baud clock according to the following formula:

$$\text{Asynchronous baud value} = \frac{\text{VCLK Frequency}}{16 * (\text{BAUD} + 1)}$$

For BAUD = 0,

$$\text{Asynchronous baud value} = \frac{\text{VCLK Frequency}}{32} \quad (57)$$

In isosynchronous timing mode, the SCI generates a baud clock according to the following formula:

$$\text{Isosynchronous baud value} = \frac{\text{VCLK Frequency}}{\text{BAUD} + 1}$$

For BAUD = 0,

$$\text{Isosynchronous baud value} = \frac{\text{VCLK Frequency}}{32} \quad (58)$$

29.2.4 SCI Multiprocessor Communication Modes

In some applications, the SCI may be connected to more than one serial communication device. In such a multiprocessor configuration, several frames of data may be sent to all connected devices or to an individual device. In the case of data sent to an individual device, the receiving devices must determine when they are being addressed. When a message is not intended for them, the devices can ignore the following data. When only two devices make up the SCI network, addressing is not needed, so multiprocessor communication schemes are not required.

SCI supports two multiprocessor Communication Modes which can be selected using COMM MODE bit:

- Idle-Line Mode
- Address Bit Mode

When the SCI is not used in a multiprocessor environment, software can consider all frames as data frames. In this case, the only distinction between the idle-line and address-bit modes is the presence of an extra bit (the address bit) in each frame sent with the address-bit protocol.

The SCI allows full-duplex communication where data can be sent and received via the transmit and receive pins simultaneously. However, the protocol used by the SCI assumes that only one device transmits data on the same bus line at any one time. No arbitration is done by the SCI.

NOTE: Avoid Transmitting Simultaneously on the Same Serial Bus

The system designer must ensure that devices connected to the same serial bus line do not attempt to transmit simultaneously. If two devices are transmitting different data, the resulting bus conflict could damage the device..

29.2.4.1 Idle-Line Multiprocessor Modes

In idle-line multiprocessor mode, a frame that is preceded by an idle period (10 or more idle bits) is an address frame. A frame that is preceded by fewer than 10 idle bits is a data frame. Figure 29-4 illustrates the format of several blocks and frames with idle-line mode.

There are two ways to transmit an address frame using idle-line mode:

Method 1: In software, deliberately leave an idle period between the transmission of the last data frame of the previous block and the address frame of the new block.

Method 2: Configure the SCI to automatically send an idle period between the last data frame of the previous block and the address frame of the new block.

Although Method 1 is only accomplished by a delay loop in software, Method 2 can be implemented by using the transmit buffer and the TXWAKE bit in the following manner:

Step 1 : Write a 1 to the TXWAKE bit.

Step 2 : Write a dummy data value to the SCITD register. This triggers the SCI to begin the idle period as soon as the transmitter shift register is empty.

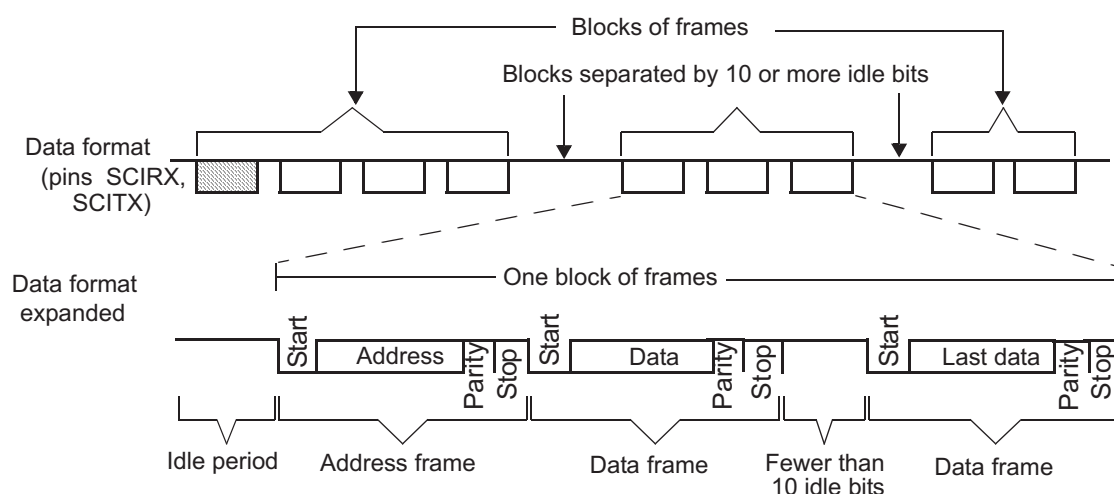
Step 3 : Wait for the SCI to clear the TXWAKE flag.

Step 4 : Write the address value to SCITD.

As indicated by Step 3, software should wait for the SCI to clear the TXWAKE bit. However, the SCI clears the TXWAKE bit at the same time it sets TXRDY (that is, transfers data from SCITD into SCITXSHF). Therefore, if the TX INT ENA bit is set, the transfer of data from SCITD to SCITXSHF causes an interrupt to be generated at the same time that the SCI clears the TXWAKE bit. If this interrupt method is used, software is not required to poll the TXWAKE bit waiting for the SCI to clear it.

When idle-line multiprocessor communications are used, software must ensure that the idle time exceeds 10 bit periods before addresses (using one of the methods mentioned above), and software must also ensure that data frames are written to the transmitter quickly enough to be sent without a delay of 10 bit periods between frames. Failure to comply with these conditions will result in data interpretation errors by other devices receiving the transmission.

Figure 29-4. Idle-Line Multiprocessor Communication Format



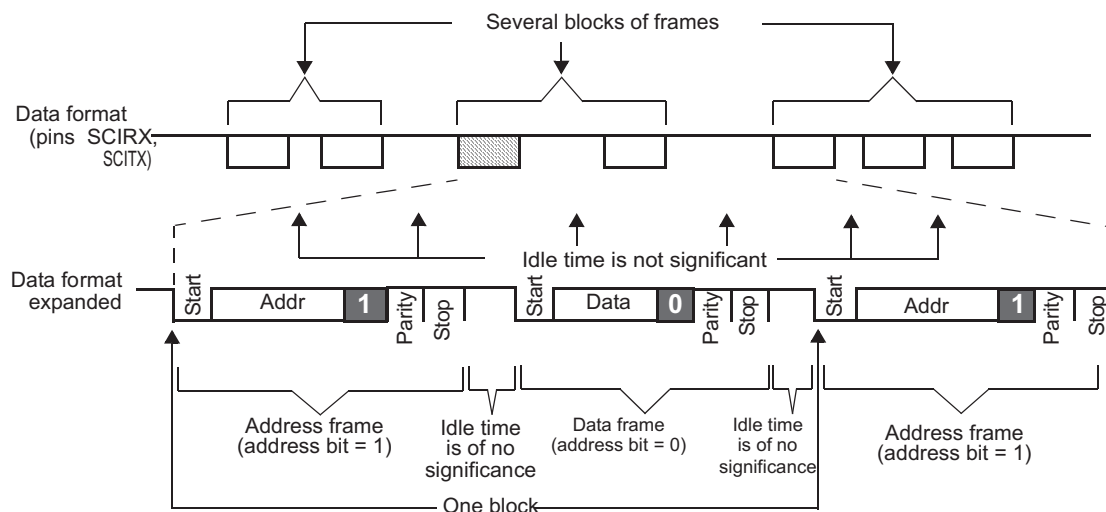
29.2.4.2 Address-Bit Multiprocessor Mode

In the address-bit protocol, each frame has an extra bit immediately following the data field called an address bit. A frame with the address bit set to 1 is an address frame; a frame with the address bit set to 0 is a data frame. The idle period timing is irrelevant in this mode. Figure 29-5 illustrates the format of several blocks and frames with the address-bit mode.

When address-bit mode is used, the value of the TXWAKE bit is the value sent as the address bit. To send an address frame, software must set the TXWAKE bit. This bit is cleared as the contents of the SCITD are shifted from the TXWAKE register so that all frames sent are data except when the TXWAKE bit is written as a 1.

No dummy write to SCITD is required before an address frame is sent in address-bit mode. The first byte written to SCITD after the TXWAKE bit is written to 1 is transmitted with the address bit set when address-bit mode is used.

Figure 29-5. Address-Bit Multiprocessor Communication Format



29.3 SCI Interrupts

The SCI module has two interrupt lines, level 0 and level 1, to the vectored interrupt manager (VIM) module (see [Figure 29-6](#)). Two offset registers SCIINTVECT0 and SCIINTVECT1 determine which flag triggered the interrupt according to the respective priority encoders. Each interrupt condition has a bit to enable and disable the interrupt in the SCISSETINT and SCICLRINT registers, respectively.

Each interrupt also has a bit that can be set as interrupt level 0 (INT0) or as interrupt level 1 (INT1). By default, interrupts are in interrupt level 0. SCISSETINTLVL sets a given interrupt to level1. SCICLEARINTLVL resets a given interrupt level to the default level 0.

The interrupt vector registers SCIINTVECT0 and SCIINTVECT1 return the vector of the pending interrupt line INT0 or INT1. If more than one interrupt is pending, the interrupt vector register holds the highest priority interrupt.

Figure 29-6. General Interrupt Scheme

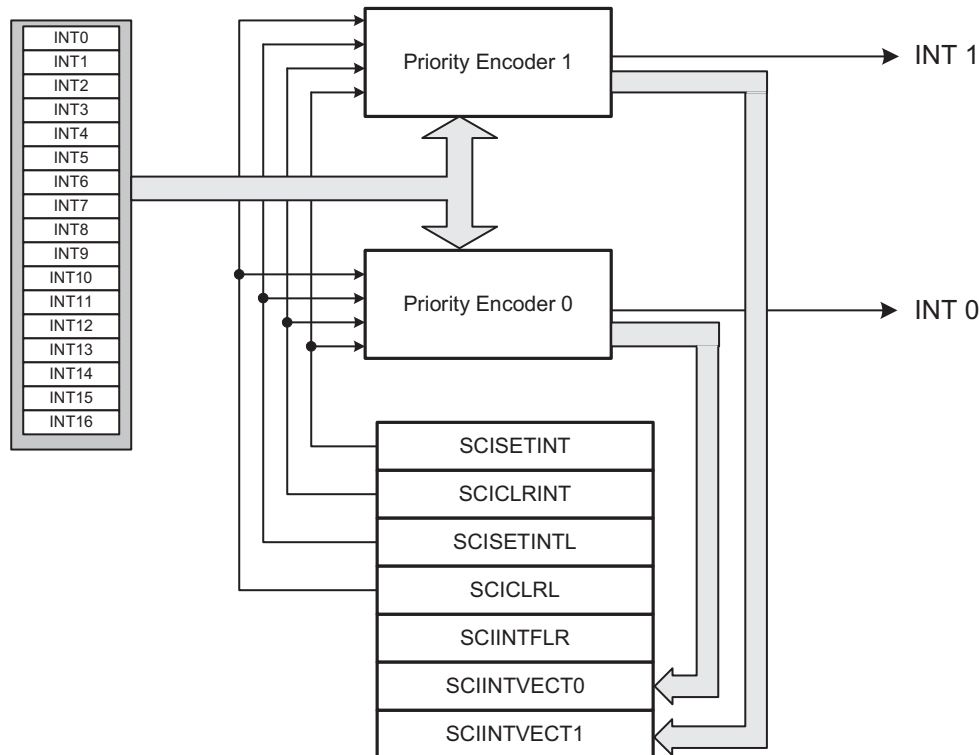
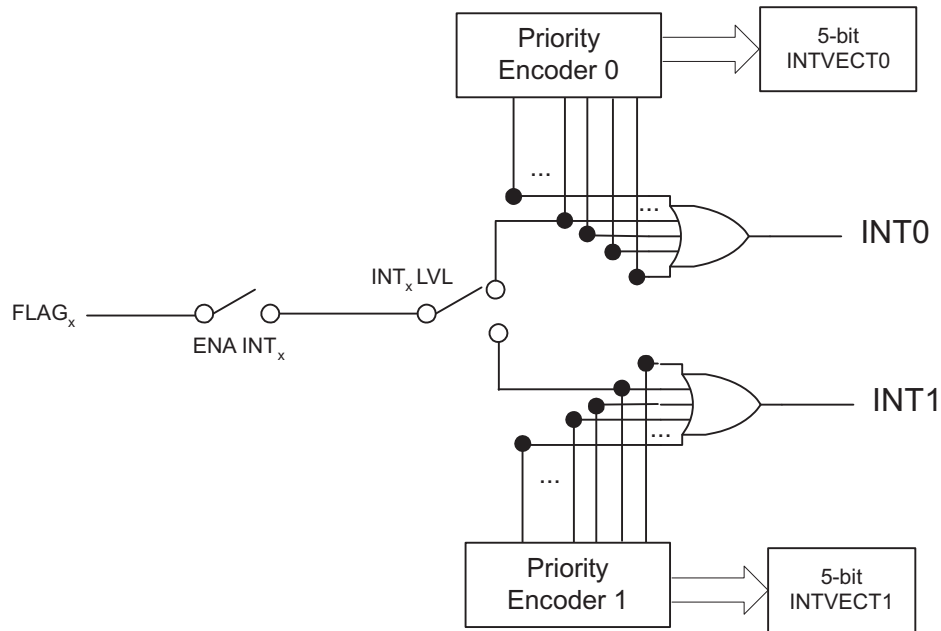


Figure 29-7. Interrupt Generation for Given Flags



29.3.1 Transmit Interrupt

To use transmit interrupt functionality, SET TX INT bit must be enabled and SET TX DMA bit must be cleared. The transmit ready (TXRDY) flag is set when the SCI transfers the contents of SCITD to the shift register, SCITXSHF. The TXRDY flag indicates that SCITD is ready to be loaded with more data. In addition, the SCI sets the TX EMPTY bit if both the SCITD and SCITXSHF registers are empty. If the SET TX INT bit is set, then a transmit interrupt is generated when the TXRDY flag goes high. Transmit Interrupt is not generated immediately after setting the SET TX INT bit unlike transmit DMA request. Transmit Interrupt is generated only after the first transfer from SCITD to SCITXSHF, that is first data has to be written to SCITD by the User before any interrupt gets generated. To transmit further data the user can write data to SCITD in the transmit Interrupt service routine.

Writing data to the SCITD register clears the TXRDY bit. When this data has been moved to the SCITXSHF register, the TXRDY bit is set again. The interrupt request can be suspended by setting the CLR TX INT bit; however, when the SET TX INT bit is again set to 1, the TXRDY interrupt is asserted again. The transmit interrupt request can be eliminated until the next series of values is written to SCITD, by disabling the transmitter via the TXENA bit, by a software reset SWnRST, or by a device hardware reset.

29.3.2 Receive Interrupt

The receive ready (RXRDY) flag is set when the SCI transfers newly received data from SCIRXSHF to SCIRD. The RXRDY flag therefore indicates that the SCI has new data to be read. Receive interrupts are enabled by the SET RX INT bit. If the SET RX INT is set when the SCI sets the RXRDY flag, then a receive interrupt is generated. The received data can be read in the Interrupt Service routine.

On a device with both SCI and a DMA controller, the bits SET RX DMA ALL and SET RX DMA must be cleared to select interrupt functionality.

29.3.3 WakeUp Interrupt

SCI sets the WAKEUP flag if bus activity on the RX line either prevents power-down mode from being entered, or RX line activity causes an exit from power-down mode. If enabled (SET WAKEUP INT), wakeup interrupt is triggered once WAKEUP flag is set.

29.3.4 Error Interrupts

The following error detection features are supported with Interrupt by the SCI module:

- Parity errors (PE)
- Frame errors (FE)
- Break Detect errors (BRKDT)
- Overrun errors (OE)

If any of these errors (PE, FE, BRKDT, OE) is flagged, an interrupt for the flagged errors will be generated if enabled. A message is valid for both the transmitter and the receiver if there is no error detected until the end of the frame. Each of these flags is located in the receiver status (SCIFLR) register. Further details on these flags are explained in SCIFLR register description.

The SCI module supports the following 7 interrupts as listed in [Table 29-1](#).

Table 29-1. SCI Interrupts

Offset ⁽¹⁾	Interrupt
0	Reserved
1	Wakeup
2	Reserved
3	Parity error
4	Reserved
5	Reserved
6	Frame error
7	Break detect error
8	Reserved
9	Overrun error
10	Reserved
11	Receive
12	Transmit
13-15	Reserved

⁽¹⁾ Offset 1 is the highest priority. Offset 16 is the lowest priority.

29.4 SCI DMA Interface

DMA requests for receive (RXDMA request) and transmit (TXDMA request) are available for the SCI module. Refer to the DMA module chapter for DMA module configurations.

29.4.1 Receive DMA Requests

This DMA functionality is enabled/disabled by the CPU using the SET RX DMA/CLR RX DMA bits, respectively.

The receiver DMA request is set when a frame is received successfully and DMA functionality has been previously enabled. The RXRDY flag is set when the SCI transfers newly received data from the SCIRXSHF register to the SCIRD buffer. The RXRDY flag therefore indicates that the SCI has new data to be read. Receive DMA requests are enabled by the SET RX INT bit.

Parity, overrun, break detect, wake-up, and framing errors generate an error interrupt request immediately upon detection, if enabled, even if the device is in the process of a DMA data transfer. The DMA transfer is postponed until the error interrupt is served. The error interrupt can delete this particular DMA request by reading the receive buffer.

In multiprocessor mode, the SCI can generate receiver interrupts for address frames and DMA requests for data frames. This is controlled by an extra select bit SET RX DMA ALL.

If the SET RX DMA ALL bit is set and the SET RX DMA bit is set when the SCI sets the RXRDY flag, then a receive DMA request is generated for address and data frames.

If the SET RX DMA ALL bit is cleared and the SET RX DMA bit is set when the SCI sets the RXRDY flag upon receipt of a data frame, then a receive DMA request is generated. Receive interrupt requests are generated for address frames.

In multiprocessor mode with the SLEEP bit set, no DMA is generated for received data frames. The software must clear the SLEEP bit before data frames can be received. [Table 29-2](#) specifies the bit values for DMA requests in multiprocessor modes.

Table 29-2. DMA and Interrupt Requests in Multiprocessor Modes

SET RX INT	SET RX DMA	SET RX DMA ALL	ADDR FRAME INT	ADDR FRAME DMA	DATA FRAME INT	DATA FRAME DMA
0	0	x	N	N	N	N
0	1	0	Y	N	N	Y
0	1	1	N	Y	N	Y
1	0	x	Y	N	Y	N
1	1	0	Y	N	Y	Y
1	1	1	Y	Y	Y	Y

In multiprocessor mode, the SCI can generate receiver interrupts for address frames and DMA requests for data frames or DMA requests for both. This is controlled by the SET RX DMA ALL bit.

In multiprocessor mode with the SLEEP bit set, no DMA is generated for received data frames. The software must clear the SLEEP bit before data frames can be received.

29.4.2 Transmit DMA Requests

DMA functionality is enabled and disabled by the CPU with the SET TX DMA and CLR TX DMA bits, respectively.

The TXRDY flag is set when the SCI transfers the contents of SCITD to SCITXSHF. The TXRDY flag indicates that SCITD is ready to be loaded with more data. In addition, the SCI sets the TX EMPTY bit if both the SCITD and SCITXSHF registers are empty.

Transmit DMA requests are enabled by the setting SET TX DMA and SET TX INT bits. If the SET TX DMA bit is set, then a TX DMA request is sent to the DMA when data is written to SCITD and TXRDY is set. The DMA will write the first byte to the transmit buffer.

29.5 SCI Configurations

Before the SCI sends or receives data, its registers should be properly configured. Upon power-up or a system-level reset, each bit in the SCI registers is set to a default state. The registers are writable only after the RESET bit in the SCIGCR0 register is set to 1. Of particular importance is the SWnRST bit in the SCIGCR1 register. The SWnRST is an active-low bit initialized to 0 and keeps the SCI in a reset state until it is programmed to 1. Therefore, all SCI configuration should be completed before a 1 is written to the SWnRST bit.

The following list details the configuration steps that software should perform prior to the transmission or reception of data. As long as the SWnRST bit is cleared to 0 the entire time that the SCI is being configured, the order in which the registers are programmed is not important.

- Enable SCI by setting the RESET bit to 1.
- Clear the SWnRST bit to 0 before SCI is configured.
- Select the desired frame format by programming the SCIGCR1 register.
- Set both the RX FUNC and TX FUNC bits in SCIPIO0 to 1 to configure the SCIRX and SCITX pins for SCI functionality.
- Select the baud rate to be used for communication by programming the BRS register.
- Set the CLOCK bit in SCIGCR1 to 1 to select the internal clock.
- Set the CONT bit in SCIGCR1 to 1 to make SCI not halt for an emulation breakpoint until its current reception or transmission is complete (this bit is used only in an emulation environment).
- Set LOOP BACK bit in SCIGCR1 to 1 to connect the transmitter to the receiver internally (this feature is used to perform a self-test).
- Set the RXENA bit in SCIGCR1 to 1, if data is to be received.
- Set the TXENA bit in SCIGCR1 to 1, if data is to be transmitted.
- Set the SWnRST bit to 1 after SCI is configured.
- Perform receiving or transmitting data (see [Section 29.5.1](#) and [Section 29.5.2](#)).

29.5.1 Receiving Data

The SCI receiver is enabled to receive messages if both the RX FUNC bit and the RXENA bit are set to 1. If the RX FUNC bit is not set, the SCIRX pin functions as a general-purpose I/O pin rather than as an SCI function pin. After a valid idle period is detected, data is automatically received as it arrives on the SCIRX pin.

SCI sets the RXRDY bit when it transfers newly received data from SCIRXSHF to SCIRD. The SCI clears the RXRDY bit after the new data in SCIRD has been read. Also, as data is transferred from SCIRXSHF to SCIRD, the SCI sets the FE, OE, or PE flags if any of these error conditions were detected in the received data. These error conditions are supported with configurable interrupt capability. The wake-up and break-detect status bits are also set if one of these errors occurs, but they do not necessarily occur at the same time that new data is being loaded into SCIRD.

You can receive data by:

1. Polling Receive Ready Flag
2. Receive Interrupt
3. DMA

In polling method, software can poll for the RXRDY bit and read the data from SCIRD register once RXRDY is set high. The CPU is unnecessarily overloaded by selecting the polling method. To avoid this, you can use either the interrupt or DMA method. To use the interrupt method, the SET RX INT bit is set. To use the DMA method, the SET RX DMA bit is set. Either an interrupt or a DMA request is generated the moment the RXRDY bit is set.

29.5.2 Transmitting Data

The SCI transmitter is enabled if both the TX FUNC bit and the TXENA bit are set to 1. If the TX FUNC bit is not set, the SCITX pin functions as a general-purpose I/O pin rather than as an SCI function pin. Any value written to the SCITD before TXENA is set to 1 is not transmitted. Both of these control bits allow for the SCI transmitter to be held inactive independently of the receiver.

SCI waits for data to be written to SCITD, transfers it to SCITXSHF, and transmits the data. The TXRDY and TX EMPTY bits indicate the status of the transmit buffers. That is, when the transmitter is ready for data to be written to SCITD, the TXRDY bit is set. Additionally, if both SCITD and SCITXSHF are empty, then the TX EMPTY bit is also set.

You can transmit data by:

1. Polling Transmit Ready Flag
2. Transmit Interrupt
3. DMA

In polling method, software can poll for the TXRDY bit to go high before writing the data to the SCITD register. The CPU is unnecessarily overloaded by selecting the polling method. To avoid this, you can use either the interrupt or DMA method. To use the interrupt method, the SET TX INT bit is set. To use the DMA method, the SET TX DMA bit is set. Either an interrupt or a DMA request is generated the moment the TXRDY bit is set. When the SCI has completed transmission of all pending frames, the SCITXSHF register and SCITD are empty, the TXRDY bit is set, and an interrupt/DMA request is generated, if enabled. Because all data has been transmitted, the interrupt/DMA request should be halted. This can either be done by disabling the transmit interrupt (CLR TX INT) / DMA request (CLR TX DMA bit) or by disabling the transmitter (clear TXENA bit).

NOTE: The TXRDY flag cannot be cleared by reading the corresponding interrupt offset in the SCIINTVECT0 or SCIINTVECT1 register.

29.6 SCI Low-Power Mode

The SCI can be put in either local or global low-power mode. Global low-power mode is asserted by the system and is not controlled by the SCI. During global low-power mode, all clocks to the SCI are turned off so the module is completely inactive.

Local low-power mode is asserted by setting the POWERDOWN bit; setting this bit stops the clocks to the SCI internal logic and the module registers. Setting the POWERDOWN bit causes the SCI to enter local low-power mode and clearing the POWERDOWN bit causes SCI to exit from local low-power mode. All the registers are accessible during local power-down mode as any register access enables the clock to SCI for that particular access alone.

The wake-up interrupt is used to allow the SCI to exit low-power mode automatically when a low level is detected on the SCIRX pin and also this clears the POWERDOWN bit. If wake-up interrupt is disabled, then the SCI immediately enters low-power mode whenever it is requested and also any activity on the SCIRX pin does not cause the SCI to exit low-power mode.

NOTE: Enabling Local Low-Power Mode During Receive and Transmit

If the wake-up interrupt is enabled and low-power mode is requested while the receiver is receiving data, then the SCI immediately generates a wake-up interrupt to clear the power-down bit and prevents the SCI from entering low-power mode and thus completes the current reception. Otherwise, if the wake-up interrupt is disabled, then the SCI completes the current reception and then enters the low-power mode.

29.6.1 Sleep Mode for Multiprocessor Communication

When the SCI receives data and transfers that data from SCIRXSHF to SCIRD, the RXRDY bit is set and if RX INT ENA is set, the SCI also generates an interrupt. The interrupt triggers the CPU to read the newly received frame before another one is received. In multiprocessor communication modes, this default behavior may be enhanced to provide selective indication of new data. When SCI receives an address frame that does not match its address, the device can ignore the data following this non-matching address until the next address frame by using sleep mode. Sleep mode can be used with both idle-line and address-bit multiprocessor modes.

If sleep mode is enabled by the SLEEP bit, then the SCI transfers data from SCIRXSHF to SCIRD only for address frames. Therefore, in sleep mode, all data frames are assembled in the SCIRXSHF register without being shifted into the SCIRD and without initiating a receive interrupt or DMA request. Upon reception of an address frame, the contents of the SCIRXSHF are moved into SCIRD, and the software must read SCIRD and determine if the SCI is being addressed by comparing the received address against the address previously set in the software and stored somewhere in memory (the SCI does not have hardware available for address comparison). If the SCI is being addressed, the software must clear the SLEEP bit so that the SCI will load SCIRD with the data of the data frames that follow the address frame.

When the SCI has been addressed and sleep mode has been disabled (in software) to allow the receipt of data, the SCI should check the RXWAKE bit (SCIFLR.12) to determine when the next address has been received. This bit is set to 1 if the current value in SCIRD is an address and set to 0 if SCIRD contains data. If the RXWAKE bit is set, then software should check the address in SCIRD against its own address. If it is still being addressed, then sleep mode should remain disabled. Otherwise, the SLEEP bit should be set again.

Following is a sequence of events typical of sleep mode operation:

- The SCI is configured and both sleep mode and receive actions are enabled.
- An address frame is received and a receive interrupt is generated.
- Software compares the received address frame against that set by software and determines that the SCI is not being addressed, so the value of the SLEEP bit is not changed.
- Several data frames are shifted into SCIRXSHF, but no data is moved to SCIRD and no receive interrupts are generated.
- A new address frame is received and a receive interrupt is generated.
- Software compares the received address frame against that set by software and determines that the SCI is being addressed and clears the SLEEP bit.
- Data shifted into SCIRXSHF is transferred to SCIRD, and a receive interrupt is generated after each data frame is received.
- In each interrupt routine, software checks RXWAKE to determine if the current frame is an address frame.
- Another address frame is received, RXWAKE is set, software determines that the SCI is not being addressed and sets the SLEEP bit back to 1. No receive interrupts are generated for the data frames following this address frame.

By ignoring data frames that are not intended for the device, fewer interrupts are generated. These interrupts would otherwise require CPU intervention to read data that is of no significance to this specific device. Using sleep mode can help free some CPU resources.

Except for the RXRDY flag, the SCI continues to update the receiver status flags (see [Table 29-11](#)) while sleep mode is active. In this way, if an error occurs on the receive line, an application can immediately respond to the error and take the appropriate corrective action.

Because the RXRDY bit is not updated for data frames when sleep mode is enabled, the SCI can enable sleep mode and use a polling algorithm if desired. In this case, when RXRDY is set, software knows that a new address has been received. If the SCI is not being addressed, then the software should not change the value of the SLEEP bit and should continue to poll RXRDY.

29.7 SCI Control Registers

These registers are accessible in 8-, 16-, and 32-bit reads or writes. The SCI is controlled and accessed through the registers listed in [Table 29-3](#). Among the features that can be programmed are the SCI communication and timing modes, baud rate value, frame format, DMA requests, and interrupt configuration. The base address for the control registers is FFF7 E500h.

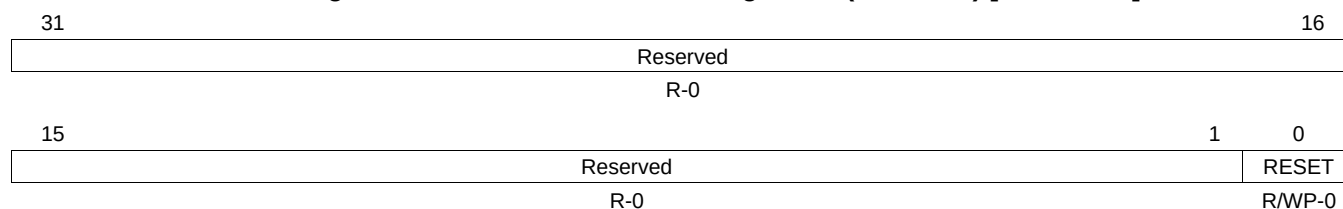
Table 29-3. SCI Control Registers Summary

Offset	Acronym	Register Description	Section
00h	SCIGCR0	SCI Global Control Register 0	Section 29.7.1
04h	SCIGCR1	SCI Global Control Register 1	Section 29.7.2
0Ch	SCISSETINT	SCI Set Interrupt Register	Section 29.7.3
10h	SCICLEARINT	SCI Clear Interrupt Register	Section 29.7.4
14h	SCISSETINTLV	SCI Set Interrupt Level Register	Section 29.7.5
18h	SCICLEARINTLV	SCI Clear Interrupt Level Register	Section 29.7.6
1Ch	SCIFLR	SCI Flags Register	Section 29.7.7
20h	SCIINTVECT0	SCI Interrupt Vector Offset 0	Section 29.7.8
24h	SCIINTVECT1	SCI Interrupt Vector Offset 1	Section 29.7.9
28h	SCIFORMAT	SCI Format Control Register	Section 29.7.10
2Ch	BRS	Baud Rate Selection Register	Section 29.7.11
30h	SCIED	Receiver Emulation Data Buffer	Section 29.7.12.1
34h	SCIRD	Receiver Data Buffer	Section 29.7.12.2
38h	SCITD	Transmit Data Buffer	Section 29.7.12.3
3Ch	SCIPIO0	SCI Pin I/O Control Register 0	Section 29.7.13
40h	SCIPIO1	SCI Pin I/O Control Register 1	Section 29.7.14
44h	SCIPIO2	SCI Pin I/O Control Register 2	Section 29.7.15
48h	SCIPIO3	SCI Pin I/O Control Register 3	Section 29.7.16
4Ch	SCIPIO4	SCI Pin I/O Control Register 4	Section 29.7.17
50h	SCIPIO5	SCI Pin I/O Control Register 5	Section 29.7.18
54h	SCIPIO6	SCI Pin I/O Control Register 6	Section 29.7.19
58h	SCIPIO7	SCI Pin I/O Control Register 7	Section 29.7.20
5Ch	SCIPIO8	SCI Pin I/O Control Register 8	Section 29.7.21
90h	IODFTCTRL	Input/Output Error Enable Register	Section 29.7.22

29.7.1 SCI Global Control Register 0 (SCIGCR0)

The SCIGCR0 register defines the module reset. [Figure 29-8](#) and [Table 29-4](#) illustrate this register.

Figure 29-8. SCI Global Control Register 0 (SCIGCR0) [offset = 00]



LEGEND: R = Read only; R/WP = Read/Write in privileged mode only; -n = value after reset

Table 29-4. SCI Global Control Register 0 (SCIGCR0) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	RESET	0 1	This bit resets the SCI module. SCI module is in reset. SCI module is out of reset. Note: Read/Write in privileged mode only.

29.7.2 SCI Global Control Register 1 (SCIGCR1)

The SCIGCR1 register defines the frame format, protocol, and communication mode used by the SCI. [Figure 29-9](#) and [Table 29-5](#) illustrate this register.

NOTE: The SCIGCR1 Control Register Bits should not be changed during Frame Transmission or Reception.

Figure 29-9. SCI Global Control Register 1 (SCIGCR1) [offset = 04h]

31	Reserved				26	25	24
	R-0					TXENA	RXENA
						R/W-0	R/W-0
23	Reserved				18	17	16
	R-0					CONT	LOOP BACK
						R/W-0	R/W-0
15	Reserved				10	9	8
	R-0					POWERDOWN	SLEEP
						R/WP-0	R/W-0
7	6	5	4	3	2	1	0
SWnRST	Reserved	CLOCK	STOP	PARITY	PARITY ENA	TIMING MODE	COMM MODE
R/W-0	R-0	R/W-0	R/WC-0	R/WC-0	R/W-0	R/WC-0	R/W-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; WC = Write in sci-compatible mode only; -n = value after reset

Table 29-5. SCI Global Control Register 1 (SCIGCR1) Field Descriptions

Bit	Field	Value	Description
31-26	Reserved	0	Reads return 0. Writes have no effect.
25	TXENA	0 1	Transmit enable. Data is transferred from SCITD to the SCITXSHF shift out register only when the TXENA bit is set. Disable transfers from SCITD to SCITXSHF. Enable SCI to transfer data from SCITD to SCITXSHF. Note: Data written to SCITD or the transmit multi-buffer before TXENA is set is not transmitted. If TXENA is cleared while transmission is ongoing, the data previously written to SCITD is sent.
24	RXENA	0 1	Receive enable. RXENA allows or prevents the transfer of data from SCIRXSHF to SCIRD. The receiver will not transfer data from the shift buffer to the receive buffer. The receiver will transfer data from the shift buffer to the receive buffer. Note: Clearing RXENA stops received characters from being transferred into the receive buffer or multi-buffers, prevents the RX status flags (see Table 29-11) from being updated by receive data, and inhibits both receive and error interrupts. However, the shift register continues to assemble data regardless of the state of RXENA. Note: If RXENA is cleared before a frame is completely received, the data from the frame is not transferred into the receive buffer. Note: If RXENA is set before a frame is completely received, the data from the frame is transferred into the receive buffer. If RXENA is set while SCIRXSHF is in the process of assembling a frame, the status flags are not assured to be accurate for that frame. To ensure that the status flags correctly reflect what was detected on the bus during a particular frame, RXENA should be set before the detection of that frame.
23-18	Reserved	0	Reads return 0. Writes have no effect.
17	CONT	0 1	Continue on suspend. This bit has an effect only when a program is being debugged with an emulator, and it determines how the SCI operates when the program is suspended. The When debug mode is entered, the SCI state machine is frozen. Transmissions are halted and resume when debug mode is exited. When debug mode is entered, the SCI continues to operate until the current transmit and receive functions are complete.

Table 29-5. SCI Global Control Register 1 (SCIGCR1) Field Descriptions (continued)

Bit	Field	Value	Description
16	LOOP BACK	0 1	Loopback bit. The self-checking option for the SCI can be selected with this bit. If the SCITX and SCIRX pins are configured with SCI functionality, then the SCITX pin is internally connected to the SCIRX pin. Externally, during loop back operation, the SCITX pin outputs a high value and the SCIRX pin is in a high-impedance state. If this bit value is changed while the SCI is transmitting or receiving data, errors may result. 0 Loop back mode is disabled. 1 Loop back mode is enabled.
15-10	Reserved	0	Reads return 0. Writes have no effect.
9	POWERDOWN	0 1	Power down. When the POWERDOWN bit is set, the SCI attempts to enter local low-power mode. If the POWERDOWN bit is set while the receiver is actively receiving data and the wake-up interrupt is enabled, then the SCI immediately asserts an error interrupt to prevent low-power mode from being entered. Only Privilege mode writes allowed. 0 Normal operation. 1 Low-power mode is enabled.
8	SLEEP	0 1	SCI sleep. In a multiprocessor configuration, this bit controls the receive sleep function. Clearing this bit brings the SCI out of sleep mode. 0 Sleep mode is disabled. 1 Sleep mode is enabled. Note: The receiver still operates when the SLEEP bit is set; however, RXRDY is updated and SCIRD is loaded with new data only when an address frame is detected. The remaining receiver status flags (see Table 29-11) are updated and an error interrupt is requested if the corresponding interrupt enable bit is set, regardless of the value of the SLEEP bit. In this way, if an error is detected on the receive data line while the SCI is asleep, software can promptly deal with the error condition. Note: The SLEEP bit is not automatically cleared when an address byte is detected. See Section 29.6 for more information on using the SLEEP bit for multiprocessor communication.
7	SWnRST	0 1	Software reset (active low). 0 The SCI is in its reset state; no data will be transmitted or received. Writing a 0 to this bit initializes the SCI state machines and operating flags as defined in Table 29-11 and Table 29-12. All affected logic is held in the reset state until a 1 is written to this bit. 1 The SCI is in its ready state; transmission and reception can be done. After this bit is set to 1, the configuration of the module should not change. Note: The SCI should only be configured while SWnRESET = 0.
6	Reserved	0	Reads return 0. Writes have no effect.
5	CLOCK	0 1	SCI internal clock enable. The CLOCK bit determines the source of the module clock on the SCICLK pin. 0 The external SCICLK is the clock source. 1 The internal SCICLK is the clock source. Note: If an external clock is selected, then the internal baud rate generator and baud rate registers are bypassed. The maximum frequency allowed for an externally sourced SCI clock is VCLK/16.
4	STOP	0 1	SCI number of stop bits per frame. 0 One stop bit is used. 1 Two stop bits are used. Note: The receiver checks for only one stop bit. However in idle-line mode, the receiver waits until the end of the second stop bit (if STOP = 1) to begin checking for an idle period.

Table 29-5. SCI Global Control Register 1 (SCIGCR1) Field Descriptions (continued)

Bit	Field	Value	Description
3	PARITY		SCI parity odd/even selection. If the PARITY ENA bit is set, PARITY designates odd or even parity.
		0	Odd parity is used.
		1	Even parity is used. The parity bit is calculated based on the data bits in each frame and the address bit (in address-bit mode). The start and stop fields in the frame are not included in the parity calculation. For odd parity, the SCI transmits and expects to receive a value in the parity bit that makes odd the total number of bits in the frame with the value of 1. For even parity, the SCI transmits and expects to receive a value in the parity bit that makes even the total number of bits in the frame with the value of 1.
2	PARITY ENA	0	Parity enable. This bit enables or disables the parity function. Parity is disabled; no parity bit is generated during transmission or is expected during reception.
		1	Parity is enabled. A parity bit is generated during transmission and is expected during reception.
1	TIMING MODE	0	SCI timing mode bit. Synchronous timing is used.
		1	Asynchronous timing is used.
0	COMM MODE		SCI communication mode bit.
		0	Idle-line mode is used.
		1	Address-bit mode is used.

29.7.3 SCI Set Interrupt Register (SCISSETINT)

Figure 29-10 and Table 29-6 illustrate this register. SCISSETINT register is used to enable the required interrupts supported by the module.

Figure 29-10. SCI Set Interrupt Register (SCISSETINT) [offset = 0Ch]

31	27	26	25	24
Reserved		SET FE INT	SET OE INT	SET PE INT
R-0		R/W-0	R/W-0	R/W-0
23	19	18	17	16
Reserved		SET RX DMA ALL	SET RX DMA	SET TX DMA
R-0		R/WC-0	R/W-0	R/W-0
15	10	9	8	
Reserved			SET RX INT	SET TX INT
R-0			R/W-0	R/W-0
7	2	1	0	
Reserved			SET WAKEUP INT	SET BRKDT INT
R-0			R/W-0	R/WC-0

LEGEND: R/W = Read/Write; R = Read only; WC = Write in sci-compatible mode only; -n = value after reset

Table 29-6. SCI Set Interrupt Register (SCISSETINT) Field Descriptions

Bit	Field	Value	Description
31-27	Reserved	0	Reads return 0. Writes have no effect.
26	SET FE INT	0	Set framing-error interrupt. Setting this bit enables the SCI module to generate an interrupt when a framing error occurs. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
25	SET OE INT	0	Set overrun-error interrupt. Setting this bit enables the SCI module to generate an interrupt when an overrun error occurs. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
24	SET PE INT	0	Set parity interrupt. Setting this bit enables the SCI module to generate an interrupt when a parity error occurs. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
23-19	Reserved	0	Reads return 0. Writes have no effect.
18	SET RX DMA ALL	0	Set receive DMA all. This bit determines if a separate interrupt is generated for the address frames sent in multiprocessor communications. When this bit is 0, RX interrupt requests are generated for address frames and DMA requests are generated for data frames. When this bit is 1, RX DMA requests are generated for both address and data frames. <i>Read:</i> The DMA request is disabled for address frames (the receive interrupt request is enabled for address frames). <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read and write:</i> The DMA request is enabled for address and data frames
17	SET RX DMA	0	Set receiver DMA. To enable receiver DMA requests, this bit must be set. If it is cleared, interrupt requests are generated depending on the SET RX INT bit (SCISSETINT[9]). <i>Read:</i> The DMA request is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read and write:</i> The DMA request is enabled for address and data frames

Table 29-6. SCI Set Interrupt Register (SCISSETINT) Field Descriptions (continued)

Bit	Field	Value	Description
16	SET TX DMA	0	Set transmit DMA. To enable DMA requests for the transmitter, this bit must be set. If it is cleared, interrupt requests are generated depending on the SET TX INT bit (SCISSETINT[8]). <i>Read:</i> Transmit DMA request is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read and write:</i> Transmit DMA request is enabled.
15-10	Reserved	0	Reads return 0. Writes have no effect.
9	SET RX INT	0	Receiver interrupt enable. Setting this bit enables the SCI to generate a receive interrupt after a frame has been completely received and the data is being transferred from SCIRXSHF to SCIRD. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
8	SET TX INT	0	Set transmitter interrupt. Setting this bit enables the SCI to generate a transmit interrupt as data is being transferred from SCITD to SCITXSHF and the TXRDY bit is being set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
7-2	Reserved	0	Reads return 0. Writes have no effect.
1	SET WAKEUP INT	0	Set wakeup interrupt. Setting this bit enables the SCI to generate a wakeup interrupt and thereby exit low-power mode. If enabled, the wakeup interrupt is asserted when local lowpower mode is requested while the receiver is busy or if a low level is detected on the SCIRX pin during low-power mode. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
0	SET BRKDT INT	0	Set break detect interrupt. Setting this bit enables the SCI to generate an error interrupt if a break condition is detected on the SCIRX pin. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.

29.7.4 SCI Clear Interrupt Register (SCICLEARINT)

Figure 29-11 and Table 29-7 illustrate this register. SCICLEARINT register is used to clear the selected enabled interrupts with out accessing SCISSETINT register.

Figure 29-11. SCI Clear Interrupt Register (SCICLEARINT) [offset = 10h]

31	27	26	25	24
Reserved		CLR FE INT	CLR OE INT	CLR PE INT
R-0		R/W-0	R/W-0	R/W-0
23	19	18	17	16
Reserved		CLR RX DMA ALL	CLR RX DMA	CLR TX DMA
R-0		R/WC-0	R/W-0	R/W-0
15	10	9	8	
Reserved			CLR RX INT	CLR TX INT
R-0			R/W-0	R/W-0
7	2	1	0	
Reserved			CLR WAKEUP INT	CLR BRKDT INT
R-0			R/W-0	R/WC-0

LEGEND: R/W = Read/Write; R = Read only; WC = Write in sci-compatible mode only; -n = value after reset

Table 29-7. SCI Clear Interrupt Register (SCICLEARINT) Field Descriptions

Bit	Field	Value	Description
31-27	Reserved	0	Reads return 0. Writes have no effect.
26	CLR FE INT	0	Clear framing-error interrupt. This bit disables the framing-error interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
25	CLR CE INT	0	Clear overrun-error interrupt. This bit disables the SCI overrun error interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
24	CLR PE INT	0	Clear parity interrupt. This bit disables the parity error interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
23-19	Reserved	0	Reads return 0. Writes have no effect.
18	CLR RX DMA ALL	0	Clear receive DMA all. This bit clears the receive DMA request for address frames when set. Only receive data frames generate a DMA request. <i>Read:</i> Receive DMA request for address frames is disabled; Instead, RX interrupt requests are enabled for address frames. Receive DMA requests are still enabled for data frames. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The receive DMA request for address and data frames is enabled. <i>Write:</i> The receive DMA request for address and data frames is disabled.

Table 29-7. SCI Clear Interrupt Register (SCICLEARINT) Field Descriptions (continued)

Bit	Field	Value	Description
17	CLR RX DMA	0	Clear receive DMA request. This bit disables the receive DMA request when set. <i>Read:</i> The DMA request is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The receive DMA request is enabled. <i>Write:</i> The receive DMA request for is disabled.
16	CLR TX DMA	0	Clear transmit DMA request. This bit disables the transmit DMA request when set. <i>Read:</i> Transmit DMA request is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The transmit DMA request is enabled. <i>Write:</i> The transmit DMA request for is disabled.
15-10	Reserved	0	Reads return 0. Writes have no effect.
9	CLR RX INT	0	Clear receiver interrupt. This bit disables the receiver interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
8	CLR TX INT	0	Clear transmitter interrupt. This bit disables the transmitter interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
7-2	Reserved	0	Reads return 0. Writes have no effect.
1	CLR WAKEUP INT	0	Clear wakeup interrupt. This bit disables the wakeup interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
0	CLR BRKDT INT	0	Clear break detect interrupt. This bit disables the break-detect interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.

29.7.5 SCI Set Interrupt Level Register (SCISSETINTLVL)

Figure 29-12 and Table 29-8 illustrate this register. This register is used to set the interrupt level for the supported interrupts.

Figure 29-12. SCI Set Interrupt Level Register (SCISSETINTLVL) [offset = 14h]

31	27	26	25	24
Reserved		SET FE INT LVL	SET OE INT LVL	SET PE INT LVL
R-0		R/W-0	R/W-0	R/W-0
23	19	18	17	16
Reserved		SET RX DMA ALL INT LVL	Reserved	
R-0		R/WC-0	R-0	
15	10	9	8	
Reserved			SET RX INT LVL	SET TX INT LVL
R-0			R/W-0	R/W-0
7	2	1	0	
Reserved			SET WAKEUP INT LVL	SET BRKDT INT LVL
R-0			R/W-0	R/WC-0

LEGEND: R/W = Read/Write; R = Read only; WC = Write in sci-compatible mode only; -n = value after reset

Table 29-8. SCI Set Interrupt Level Register (SCISSETINTLVL) Field Descriptions

Bit	Field	Value	Description
31-27	Reserved	0	Reads return 0. Writes have no effect.
26	SET FE INT LVL	0	Set framing-error interrupt level. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
25	SET CE INT LVL	0	Set overrun-error interrupt level. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
24	SET PE INT LVL	0	Set parity error interrupt level. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
23-19	Reserved	0	Reads return 0. Writes have no effect.
18	SET RX DMA ALL LVL	0	Set receive DMA all interrupt levels. <i>Read:</i> The receive interrupt request for address frames is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The receive interrupt request for address frames is mapped to the INT1 line.
17-10	Reserved	0	Reads return 0. Writes have no effect.
9	SET RX INT LVL	0	Set receiver interrupt level. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.

Table 29-8. SCI Set Interrupt Level Register (SCISSETINTLVL) Field Descriptions (continued)

Bit	Field	Value	Description
8	SET TX INT LVL	0	Set transmitter interrupt level. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
7-2	Reserved	0	Reads return 0. Writes have no effect.
1	SET WAKEUP INT LVL	0	Set wakeup interrupt level. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
0	SET BRKDT INT LVL	0	Set break detect interrupt level. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.

29.7.6 SCI Clear Interrupt Level Register (SCICLEARINTLVL)

Figure 29-13 and Table 29-9 illustrate this register.

Figure 29-13. SCI Clear Interrupt Level Register (SCICLEARINTLVL) [offset = 18h]

31	27	26	25	24
Reserved		CLR FE INT LVL	CLR OE INT LVL	CLR PE INT LVL
R-0		R/W-0	R/W-0	R/W-0
23	19	18	17	16
Reserved		CLR RX DMA ALL INT LVL	Reserved	
R-0		R/WC-0	R-0	
15	10	9	8	
Reserved			CLR RX INT LVL	CLR TX INT LVL
R-0			R/W-0	R/W-0
7	2	1	0	
Reserved			CLR WAKEUP INT LVL	CLR BRKDT INT LVL
R-0			R/W-0	R/WC-0

LEGEND: R/W = Read/Write; R = Read only; WC = Write in sci-compatible mode only; -n = value after reset

Table 29-9. SCI Clear Interrupt Level Register (SCICLEARINTLVL) Field Descriptions

Bit	Field	Value	Description
31-27	Reserved	0	Reads return 0. Writes have no effect.
26	CLR FE INT LVL	0	Clear framing-error interrupt. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.

Table 29-9. SCI Clear Interrupt Level Register (SCICLEARINTLVL) Field Descriptions (continued)

Bit	Field	Value	Description
25	CLR CE INT LVL	0	Clear overrun-error interrupt. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
24	CLR PE INT LVL	0	Clear parity interrupt. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
23-19	Reserved	0	Reads return 0. Writes have no effect.
18	CLR RX DMA ALL LVL	0	Clear receive DMA interrupt level. <i>Read:</i> The receive interrupt request for address frames is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The receive interrupt request for address frames is mapped to the INT1 line. <i>Write:</i> The receive interrupt request for address frames is mapped to the INT0 line.
17-10	Reserved	0	Reads return 0. Writes have no effect.
9	CLR RX INT LVL	0	Clear receiver interrupt. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
8	CLR TX INT LVL	0	Clear transmitter interrupt. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
7-2	Reserved	0	Reads return 0. Writes have no effect.
1	CLR WAKEUP INT LVL	0	Clear wakeup interrupt. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
0	CLR BRKDT INT LVL	0	Clear break detect interrupt. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.

29.7.7 SCI Flags Register (SCIFLR)

Figure 29-14 and Table 29-10 illustrate this register.

Figure 29-14. SCI Flags Register (SCIFLR) [offset = 1Ch]

31	27			26	25	24
Reserved				FE	OE	PE
R-0				R/W-0	R/W-0	R/W-0
23						16
Reserved						
R-0						
15	13	12	11	10	9	8
Reserved		RX WAKE	TX EMPTY	TX WAKE	RX RDY	TX RDY
R-0		R/WC-0	R/W-1	R/WC-0	R/W-0	R/W-1
7	4		3	2	1	0
Reserved			BUSY	IDLE	WAKE UP	BRKDT
R-0			R/W-0	R-0	R/WL-0	R/WC-0

LEGEND: R/W = Read/Write; R = Read only; WC = Write in sci-compatible mode only; -n = value after reset

Table 29-10. SCI Flags Register (SCIFLR) Field Descriptions

Bit	Field	Value	Description
31-27	Reserved		Reads return 0. Writes have no effect.
26	FE	0	Framing error flag. This bit is set when an expected stop bit is not found. In SCI compatibility mode, only the first stop bit is checked. The missing stop bit indicates that synchronization with the start bit has been lost and that the character is incorrectly framed. Detection of a framing error causes the SCI to generate an error interrupt, if the SET FE INT bit (SCISSETINT[26]) is set. The framing error flag is cleared by the following: - Setting of the SW nRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reading the corresponding interrupt offset in SCIINTVECT0/1 - Reception of a new character In multi-buffer mode, the frame is defined in the SCIFORMAT register. Read: No framing error has been detected since the last clear. Write: Writing a 0 to this bit has no effect.
		1	Read: A framing error has been detected since the last clear. Write: The bit is cleared to 0.
25	OE	0	Overrun error flag. This bit is set when the transfer of data from SCIRXSHF to SCIRD overwrites unread data already in SCIRD. Detection of an overrun error causes the SCI to generate an error interrupt if the SET OE INT bit (SCISSETINT[25]) is set. The OE flag is reset by the following: - Setting of the SW nRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reading the corresponding interrupt offset in SCIINTVECT0/1 Read: No overrun error has been detected since the last clear. Write: Writing a 0 to this bit has no effect.
		1	Read: An overrun error has been detected since the last clear. Write: The bit is cleared to 0.

Table 29-10. SCI Flags Register (SCIFLR) Field Descriptions (continued)

Bit	Field	Value	Description
24	PE		Parity error flag. This bit is set when a parity error is detected in the received data. In SCI address-bit mode, the parity is calculated on the data and address bit fields of the received frame. In idle-line mode, only the data is used to calculate parity. An error is generated when a character is received with a mismatch between the number of 1s and its parity bit. If the parity function is disabled (SCIGCR[2] = 0), the PE flag is disabled and read as 0. Detection of a parity error causes the SCI to generate an error interrupt if the SET PE INT bit (SCISSETINT[24]) is set. The PE bit is reset by the following: - Setting of the SW nRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reception of a new character - Reading the corresponding interrupt offset in SCIINTVECT0/1
		0	<i>Read:</i> No parity error has been detected since the last clear. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> A parity error has been detected since the last clear. <i>Write:</i> The bit is cleared to 0.
23-13	Reserved	0	Reads return 0. Writes have no effect.
12	RXWAKE		Receiver wakeup detect flag. The SCI sets this bit to indicate that the data currently in SCIRD is an address. RXWAKE is cleared by the following: - Setting of the SW nRST bit - Setting of the RESET bit - A system reset - Upon receipt of a data frame.
		0	The data in SCIRD is not an address.
		1	The data in SCIRD is an address.
11	TX EMPTY		Transmitter empty flag. This flag indicates the transmitter's buffer register(s) (SCITD/TDy) and shift register (SCITXSHF) are empty. Note: The RESET bit, an active SW nRESET (SCIGCR1[7]) or a system reset sets this bit. This bit does not cause an interrupt request.
		0	Transmitter buffer or shift register (or both) are loaded with data.
		1	Transmitter buffer and shift registers are both empty.
10	TXWAKE		Transmitter wakeup method select. The TXWAKE bit controls whether the data in SCITD should be sent as an address or data frame using multiprocessor communication format. This bit is set to 1 or 0 by software before a byte is written to SCITD and is cleared by the SCI when data is transferred from SCITD to SCITXSHF or by a system reset. Note: TXWAKE is not cleared by the SW nRESET bit.
			<i>Address-bit mode</i>
		0	Frame to be transmitted will be data (address bit = 0).
		1	Frame to be transmitted will be an address (address bit = 1).
			<i>Idle-line mode</i>
		0	The frame to be transmitted will be data.
		1	The following frame to be transmitted will be an address (writing a 1 to this bit followed by writing dummy data to the SCITD will result in a idle period of 11 bit periods before the next frame is transmitted).

Table 29-10. SCI Flags Register (SCIFLR) Field Descriptions (continued)

Bit	Field	Value	Description
9	RXRDY		Receiver ready flag. The receiver sets this bit to indicate that the SCIRD contains new data and is ready to be read by the CPU or DMA. The SCI generates a receive interrupt when RXRDY flag bit is set if the SET RX INT bit (SCISSETINT[9]) is set; RXRDY is cleared by the following: - Setting of the SW nRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reading the SCIRD register in compatibility mode - Reading the last data byte RDY of the response in SCI mode Note: The RXRDY flag cannot be cleared by reading the corresponding interrupt offset in the SCIINTVECT0/1 register.
		0	<i>Read:</i> No new data is in SCIRD. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> New data is ready to be read from SCIRD. <i>Write:</i> The bit is cleared to 0.
8	TXRDY		Transmitter buffer register ready flag. When set, this bit indicates that the transmit buffer is ready to get another character from a CPU or DMA write. Writing data to SCITD automatically clears this bit. This bit is set after the data of the TX buffer is shifted into the SCITXSHF register. This event can trigger a transmit interrupt after data is copied to the TX shift register SCITXSHF, if the SET TX INT bit (SCISSETINT[8]) is set. Note: 1) TXRDY is also set to 1 by setting of the RESET bit, setting of the RESET bit, or by a system reset. 2) The TXRDY flag cannot be cleared by reading the corresponding interrupt offset in the SCIINTVECT0/1 register. 3) The transmit interrupt request can be eliminated until the next series of data written into the transmit buffers SCITD0 and SCITD1, by disabling the corresponding interrupt via the SCICLEARINT register or by disabling the transmitter via the TXENA bit (SCIGCR1[25]).
		0	SCITD is full.
		1	SCITD is ready to receive the next character.
7-4	Reserved	0	Reads return 0. Writes have no effect.
3	BUSY		Bus busy flag. This bit indicates whether the receiver is in the process of receiving a frame. As soon as the receiver detects the beginning of a start bit, the BUSY bit is set to 1. When the reception of a frame is complete, the SCI clears the BUSY bit. If the SET WAKEUP INT bit (SCISSETINT[2]) is set and power down is requested while this bit is set, the SCI automatically prevents low-power mode from being entered and generates wakeup interrupt. The BUSY bit is controlled directly by the SCI receiver, but this bit can also be cleared by the following: - Setting the SW nRST bit - Setting of the RESET bit - A system reset occurring
		0	The receiver is not currently receiving a frame.
		1	The receiver is currently receiving a frame.
2	IDLE		SCI receiver in idle state. While this bit is set, the SCI looks for an idle period to resynchronize itself with the bit stream. The receiver does not receive any data while the bit is set. The bus must be idle for 11 bit periods to clear this bit. The SCI enters the idle state if one of the following events occurs: - A system reset - An SCI software reset - A power down - The RX pin is configured as a general I/O pin
		0	The idle period has been detected; the SCI is ready to receive.
		1	The idle period has not been detected; the SCI will not receive any data.

Table 29-10. SCI Flags Register (SCIFLR) Field Descriptions (continued)

Bit	Field	Value	Description
1	WAKEUP	0	Wakeup flag. This bit is set by the SCI when receiver or transmitter activity has taken the module out of power-down mode. An interrupt is generated if the SET WAKEUP INT bit (SCISSETINT[2]) is set. It is cleared by the following: - Setting of the SW nRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reading the corresponding interrupt offset in SCIINTVECT0/1 For compatibility mode, see the SCI document for more information on low-power mode. <i>Read:</i> The module will not wake up from power-down mode. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> Wake up from power-down mode. <i>Write:</i> The bit is cleared to 0.
0	BRKDT	0	SCI break-detect flag. This bit is set when the SCI detects a break condition on the SCIRX pin. A break condition occurs when the SCIRX pin remains continuously low for at least 10 bits after a missing first stop bit, that is, after a framing error. Detection of a break condition causes the SCI to generate an error interrupt if the SET BRKDT INT bit (SCISSETINT[0]) is set. The BRKDT bit is reset by the following: - Setting of the SW nRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reading the corresponding interrupt offset in SCIINTVECT0/1 <i>Read:</i> No break condition has been detected since the last clear. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> A break condition has been detected. <i>Write:</i> The bit is cleared to 0.

Table 29-11. SCI Receiver Status Flags

SCI Flag	Register	Bit	Value After SW nRESET ⁽¹⁾
FE	SCIFLR	26	0
OE	SCIFLR	25	0
PE	SCIFLR	24	0
RXWAKE	SCIFLR	12	0
RXRDY	SCIFLR	9	0
BRKDT	SCIFLR	0	0

⁽¹⁾ The flags are frozen with their reset value while SW nRESET = 0.

Table 29-12. SCI Transmitter Status Flags

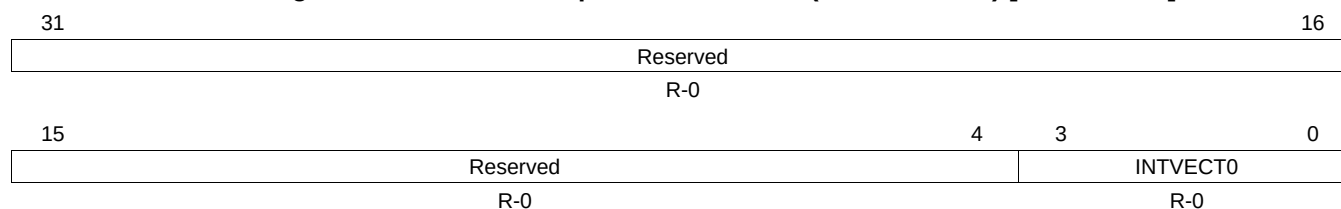
SCI Flag	Register	Bit	Value After SW nRESET ⁽¹⁾
TX EMPTY	SCIFLR	11	1
TXRDY	SCIFLR	8	1

⁽¹⁾ The flags are frozen with their reset value while SW nRESET = 0.

29.7.8 SCI Interrupt Vector Offset 0 (SCIINTVECT0)

Figure 29-15 and Table 29-13 illustrate this register.

Figure 29-15. SCI Interrupt Vector Offset 0 (SCIINTVECT0) [offset = 20h]



LEGEND: R = Read only; -n = value after reset

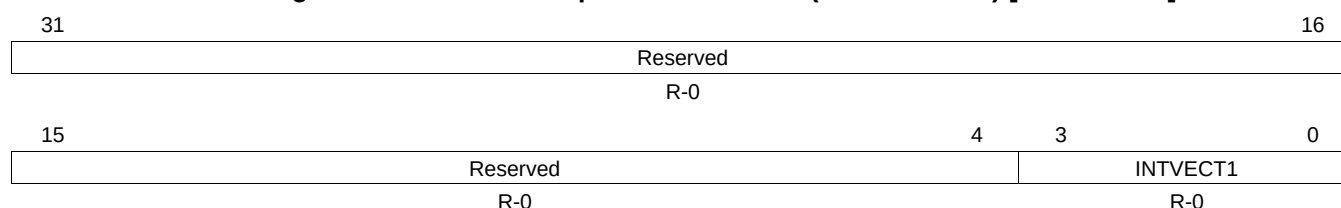
Table 29-13. SCI Interrupt Vector Offset 0 (SCIINTVECT0) Field Descriptions

Bit	Field	Value	Description
31-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	INVECT0	0-Fh	Interrupt vector offset for INT0. This register indicates the offset for interrupt line INT0. A read to this register updates its value to the next highest priority pending interrupt in SCIFLR and clears the flag in SCIFLR corresponding to the offset that was read. See Table 29-1 for a list of the interrupts. Note: The flags for the receive (SCIFLR[9]) and the transmit (SCIFLR[8]) interrupt cannot be cleared by reading the corresponding offset vector in this register (see detailed description in SCIFLR register).

29.7.9 SCI Interrupt Vector Offset 1 (SCIINTVECT1)

Figure 29-16 and Table 29-14 illustrate this register.

Figure 29-16. SCI Interrupt Vector Offset 1 (SCIINTVECT1) [offset = 24h]



LEGEND: R = Read only; -n = value after reset

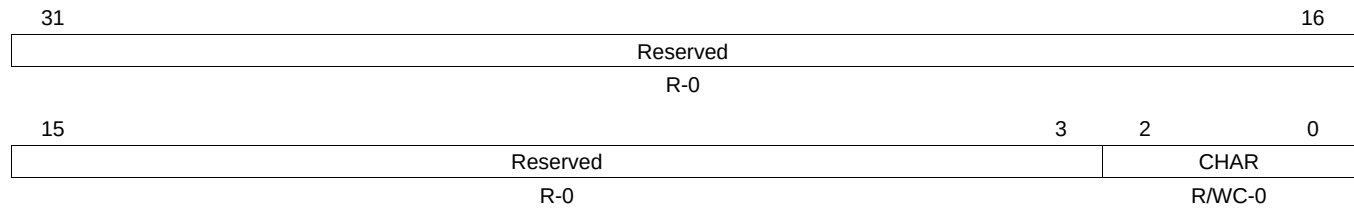
Table 29-14. SCI Interrupt Vector Offset 1 (SCIINTVECT1) Field Descriptions

Bit	Field	Value	Description
31-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	INVECT1	0-Fh	Interrupt vector offset for INT1. This register indicates the offset for interrupt line INT1. A read to this register updates its value to the next highest priority pending interrupt in SCIFLR and clears the flag in SCIFLR corresponding to the offset that was read. See Table 29-1 for list of interrupts. Note: The flags for the receive (SCIFLR[9]) and the transmit (SCIFLR[8]) interrupt cannot be cleared by reading the corresponding offset vector in this register (see detailed description in SCIFLR register).

29.7.10 SCI Format Control Register (SCIFORMAT)

Figure 29-17 and Table 29-15 illustrate this register.

Figure 29-17. SCI Format Control Register (SCIFORMAT) [offset = 28h]



LEGEND: R/W = Read/Write; R = Read only; WC = Write in SCI-compatible mode only; -n = value after reset

Table 29-15. SCI Format Control Register (SCIFORMAT) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2-0	CHAR		Character length control bits. These bits set the SCI character length from 1 to 8 bits. When data of fewer than eight bits in length is received, it is left-justified in SCIRD and padded with trailing zeros. Data read from the SCIRD should be shifted by software to make the received data right-justified. Data written to the SCITD should be right-justified but does not need to be padded with leading zeros.
		0	The character is 1 bit long.
		1h	The character is 2 bits long.
		2h	The character is 3 bits long.
		3h	The character is 4 bits long.
		4h	The character is 5 bits long.
		5h	The character is 6 bits long.
		6h	The character is 7 bits long.
		7h	The character is 8 bits long.

29.7.11 Baud Rate Selection Register (BRS)

This section describes the baud rate selection register. [Figure 29-18](#) and [Table 29-16](#) illustrate this register.

Figure 29-18. Baud Rate Selection Register (BRS) [offset = 2Ch]

31	24	23	16
Reserved		BAUD	
R-0		R/W-0	
15			0
BAUD			
R/W-0			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 29-16. Baud Rate Selection Register (BRS) Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-0	BAUD	0-FF FFFFh	SCI 24-bit baud selection. The SCI has an internally-generated serial clock determined by the VCLK and the prescalers BAUD in this register. The SCI uses the 24-bit integer prescaler BAUD value of this register to select one of over 16,700,000. The baud rate can be calculated using the following formulas: $\text{Asynchronous baud value} = \left(\frac{\text{VCLK Frequency}}{16(\text{Baud} + 1)} \right) \quad (59)$ $\text{Isosynchronous baud value} = \left(\frac{\text{VCLK Frequency}}{\text{Baud} + 1} \right) \quad (60)$ For BAUD = 0, $\text{Asynchronous baud value} = \left(\frac{\text{VCLK Frequency}}{32} \right) \quad (61)$ $\text{Isosynchronous baud value} = \left(\frac{\text{VCLK Frequency}}{2} \right) \quad (62)$ Table 29-17 contains comparative baud values for different P values, with VCLK = 50 MHz, for asynchronous mode.

Table 29-17. Comparative Baud Values (Asynchronous Mode) ⁽¹⁾⁽²⁾

24-Bit Register Value		Baud Selected		Percent Error
Decimal	Hex	Ideal	Actual	
26	00001A	115200	115740	0.47
53	000035	57600	57870	0.47
80	000050	38400	38580	0.47
162	0000A2	19200	19172	-0.15
299	00012B	10400	10417	0.16
325	000145	9600	9586	-0.15
399	00018F	7812.5	7812.5	0.00
650	00028A	4800	4800	0.00
15624	003BA0	200	200	0.00
624999	098967	5	5	0.00

⁽¹⁾ VCLK = 50 MHz

⁽²⁾ Values are in decimal except for column 2.

29.7.12 SCI Data Buffers (SCIED, SCIRD, SCITD)

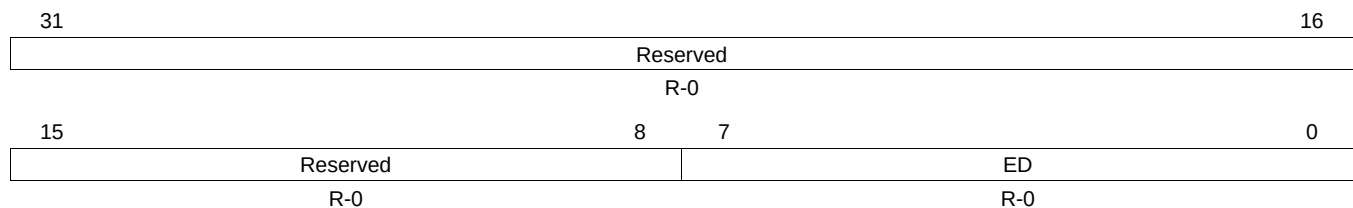
The SCI has three addressable registers in which transmit and receive data is stored.

29.7.12.1 Receiver Emulation Data Buffer (SCIED)

The SCIED register is addressed at a location different from SCIRD, but is physically the same register.

Figure 29-19 and Table 29-18 illustrate this register.

Figure 29-19. Receiver Emulation Data Buffer (SCIED) [offset = 30h]



LEGEND: R = Read only; -n = value after reset

Table 29-18. Receiver Emulation Data Buffer (SCIED) Field Descriptions

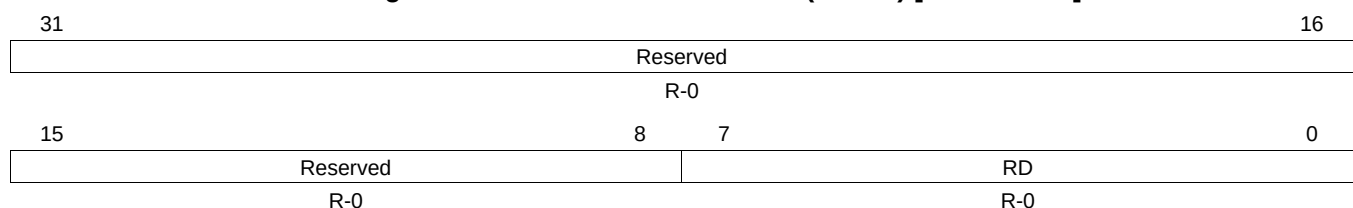
Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	ED	0-FFh	Emulator data. Reading SCIED[7:0] does not clear the RXRDY flag, unlike reading SCIRD. This register should be used only by an emulator that must continually read the data buffer without affecting the RXRDY flag.

29.7.12.2 Receiver Data Buffer (SCIRD)

This register provides a location for the receiver data. Figure 29-20 and Table 29-19 illustrate this register.

NOTE: When the SCI receives data that is fewer than eight bits in length, it loads the data into this register in a left-justified format padded with trailing zeros. Therefore, the user software should perform a logical shift on the data by the correct number of positions to make it right justified.

Figure 29-20. Receiver Data Buffer (SCIRD) [offset = 34h]



LEGEND: R = Read only; -n = value after reset

Table 29-19. Receiver Data Buffer (SCIRD) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	RD	0-FFh	Receiver data. When a frame has been completely received, the data in the frame is transferred from the receiver shift register SCIRXSHF to this register. As this transfer occurs, the RXRDY flag is set and a receive interrupt is generated if SET RX INT is set. Note: When the data is read from SCIRD, the RXRDY flag is automatically cleared.

29.7.12.3 Transmit Data Buffer Register (SCITD)

Data to be transmitted is written to the SCITD register. [Figure 29-21](#) and [Table 29-20](#) illustrate this register.

NOTE: Data written to the SCITD register that is fewer than eight bits long must be right justified, but it does not need to be padded with leading zeros.

Figure 29-21. Transmit Data Buffer Register (SCITD) [offset = 38h]

31	Reserved																16
R-0																	
15	Reserved								8	7	TD						0
R-0										R/W-0							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 29-20. Transmit Data Buffer Register (SCITD) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	TD	0-FFh	Transmit data. Data to be transmitted is written to the SCITD register. The transfer of data from this register to the transmit shift register SCITXSHF sets the TXRDY flag, which indicates that SCITD is ready to be loaded with another byte of data. Note: If TX INT ENA is set, this data transfer also causes an interrupt.

29.7.13 SCI Pin I/O Control Register 0 (SCIPIO0)

[Figure 29-22](#) and [Table 29-21](#) illustrate this register.

Figure 29-22. SCI Pin I/O Control Register 0 (SCIPIO0) [offset = 3Ch]

31											8
Reserved											
R-0											
7						3	2	1	0		
Reserved						TX FUNC		RX FUNC		Reserved	
R-0						R/W-0		R/W-0		R/W-0	

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 29-21. SCI Pin I/O Control Register 0 (SCIPIO0) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX FUNC	0 1	Transfer function. This bit defines the function of pin SCITX. SCITX is a general-purpose digital I/O pin. SCITX is the SCI transmit pin.
1	RX FUNC	0 1	Receive function. This bit defines the function of pin SCIRX. SCIRX is a general-purpose digital I/O pin. SCIRX is the SCI receive pin.
0	Reserved	0	Reads return 0. Writes have no effect.

29.7.14 SCI Pin I/O Control Register 1 (SCIPIO1)

Figure 29-23 and Table 29-22 illustrate this register.

Figure 29-23. SCI Pin I/O Control Register 1 (SCIPIO1) [offset = 40h]

31																													8
Reserved																													
R-0																													
7															3	2	1	0											
Reserved															TX DIR		RX DIR		Reserved										
R-0															R/W-0		R/W-0		R/W-0										

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 29-22. SCI Pin I/O Control Register 1 (SCIPIO1) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX DIR	0 1	Transmit pin direction. This bit determines the data direction on the SCITX pin if it is configured with general-purpose I/O functionality (TX FUNC = 0). See Table 29-23 for the SCITX pin control with this bit and others. SCITX is a general-purpose input pin. SCITX is a general-purpose output pin.
1	RX DIR	0 1	Receive pin direction. This bit determines the data direction on the SCIRX pin if it is configured with general-purpose I/O functionality (RX FUNC = 0). See Table 29-24 for the SCIRX pin control with this bit and others. SCIRX is a general-purpose input pin. SCIRX is a general-purpose output pin.
0	Reserved	0	Reads return 0. Writes have no effect.

Table 29-23. SCITX Pin Control

Function	TX IN ⁽¹⁾	TX OUT	TX FUNC	TX DIR
SCITX	X	X	1	X
General-purpose input	X	X	0	0
General-purpose output, high	X	1	0	1
General-purpose output, low	X	0	0	1

⁽¹⁾ TX IN is a read-only bit. Its value always reflects the level of the SCITX pin.

Table 29-24. SCIRX Pin Control

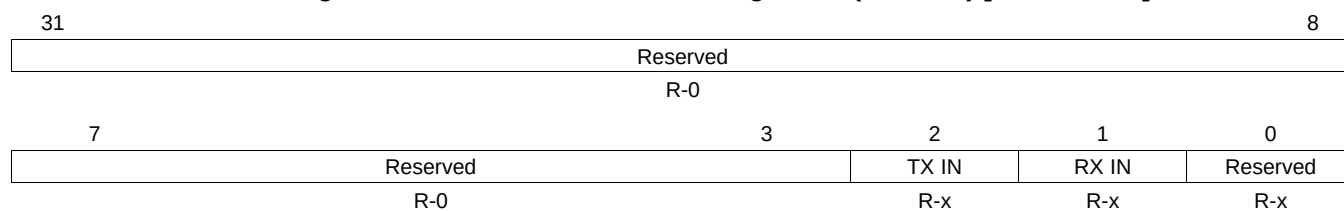
Function	RX IN ⁽¹⁾	RX OUT	RX FUNC	RX DIR
SCIRX	X	X	1	X
General-purpose input	X	X	0	0
General-purpose output, high	X	1	0	1
General-purpose output, low	X	0	0	1

⁽¹⁾ RX IN is a read-only bit. Its value always reflects the level of the SCIRX pin.

29.7.15 SCI Pin I/O Control Register 2 (SCIPIO2)

Figure 29-24 and Table 29-25 illustrate this register.

Figure 29-24. SCI Pin I/O Control Register 2 (SCIPIO2) [offset = 44h]



LEGEND: R = Read only; -n = value after reset; -x = Indeterminate

Table 29-25. SCI Pin I/O Control Register 2 (SCIPIO2) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX IN	0 1	Transmit pin in. This bit contains the current value on the SCITX pin. The SCITX pin is at logic low (0). The SCITX pin is at logic high (1).
1	RX IN	0 1	Receive pin in. This bit contains the current value on the SCIRX pin. The SCIRX pin is at logic low (0). The SCIRX pin is at logic high (1).
0	Reserved	0	Writes have no effect.

29.7.16 SCI Pin I/O Control Register 3 (SCIPIO3)

Figure 29-25 and Table 29-26 illustrate this register.

Figure 29-25. SCI Pin I/O Control Register 3 (SCIPIO3) [offset = 48h]

31																													8
Reserved																													
R-0																													
7															3	2	1	0											
Reserved															TX OUT		RX OUT		Reserved										
R-0															R/W-0		R/W-0		R/W-0										

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 29-26. SCI Pin I/O Control Register 3 (SCIPIO3) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX OUT	0 The output on the SCITX is at logic low (0). 1 The output on the SCITX pin is at logic high (1). (Output voltage is V_{OH} or higher if TXPDR = 0 and output is in high impedance state if TXPDR = 1)	Transmit pin out. This pin specifies the logic to be output on pin SCITX if the following conditions are met: - TX FUNC = 0 (SCITX pin is a general-purpose I/O.) - TX DIR = 1 (SCITX pin is a general-purpose output.) See Table 29-23 for an explanation of this bit's effect in combination with other bits.
1	RX OUT	0 The output on the SCIRX is at logic low (0). 1 The output on the SCIRX pin is at logic high (1). (Output voltage is V_{OH} or higher if RXPDR = 0, and output is in high impedance state if RXPDR = 1)	Receive pin out. This bit specifies the logic to be output on pin SCIRX if the following conditions are met: - RX FUNC = 0 (SCIRX pin is a general-purpose I/O.) - RX DIR = 1 (SCIRX pin is a general-purpose output.) See Table 29-24 for an explanation of this bit's effect in combination with the other bits.
0	Reserved	0	Reads return 0. Writes have no effect.

29.7.17 SCI Pin I/O Control Register 4 (SCIPIO4)

Figure 29-26 and Table 29-27 illustrate this register.

Figure 29-26. SCI Pin I/O Control Register 4 (SCIPIO4) [offset = 4Ch]

31																													8
Reserved																													
R-0																													
7															3	2	1	0											
Reserved															TX SET		RX SET		Reserved										
R-0															R/W-0		R/W-0		R/W-0										

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

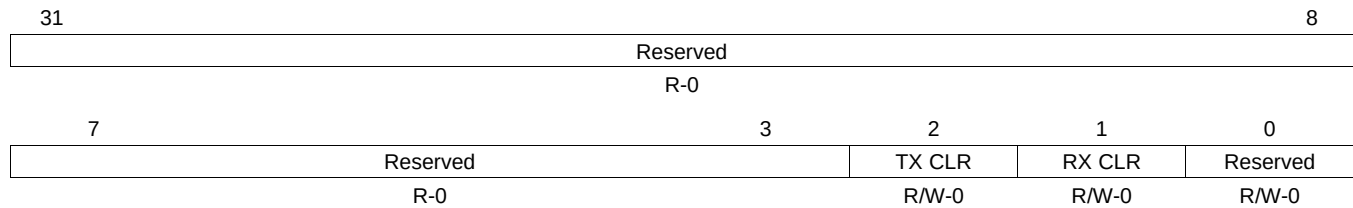
Table 29-27. SCI Pin I/O Control Register 4 (SCIPIO4) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX SET	0 1	Transmit pin set. This bit sets the logic to be output on pin SCITX if the following conditions are met: - TX FUNC = 0 (SCITX pin is a general-purpose I/O.) - TX DIR = 1 (SCITX pin is a general-purpose output.) See Table 29-23 for an explanation of this bit's effect in combination with other bits. <i>Read:</i> The output on SCITX is at logic low (0). <i>Write:</i> Writing a 0 to this bit has no effect. <i>Read and write:</i> The output on SCITX is at logic high (1).
1	RX SET	0 1	Receive pin set. This bit sets the data to be output on pin SCIRX if the following conditions are met: - RX FUNC = 0 (SCIRX pin is a general-purpose I/O.) - RX DIR = 1 (SCIRX pin is a general-purpose output.) See Table 29-24 for an explanation of this bit's effect in combination with the other bits. <i>Read:</i> The output on SCIRX is at logic low (0). <i>Write:</i> Writing a 0 to this bit has no effect. <i>Read and write:</i> The output on SCIRX is at logic high (1).
0	Reserved	0	Reads return 0. Writes have no effect.

29.7.18 SCI Pin I/O Control Register 5 (SCIPIO5)

Figure 29-27 and Table 29-28 illustrate this register.

Figure 29-27. SCI Pin I/O Control Register 5 (SCIPIO5) [offset = 50h]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 29-28. SCI Pin I/O Control Register 5 (SCIPIO5) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX CLR	0	Transmit pin clear. This bit clears the logic to be output on pin SCITX if the following conditions are met: - TX FUNC = 0 (SCITX pin is a general-purpose I/O.) - TX DIR = 1 (SCITX pin is a general-purpose output.) Read: The output on SCITX is at logic low (0). Write: Writing a 0 to this bit has no effect.
		1	Read: The output on SCITX is at logic high (1). Write: The output on SCITX is at logic low (0).
1	RX CLR	0	Receive pin clear. This bit clears the logic to be output on pin SCIRX if the following conditions are met: - RX FUNC = 0 (SCIRX pin is a general-purpose I/O.) - RX DIR = 1 (SCIRX pin is a general-purpose output.) Read: The output on SCIRX is at logic low (0). Write: Writing a 0 to this bit has no effect.
		1	Read: The output on SCIRX is at logic high (1). Write: The output on SCIRX is at logic low (0).
0	Reserved	0	Reads return 0. Writes have no effect.

29.7.19 SCI Pin I/O Control Register 6 (SCIPIO6)

Figure 29-28 and Table 29-29 illustrate this register.

Figure 29-28. SCI Pin I/O Control Register 6 (SCIPIO6) [offset = 54h]

31																													8
Reserved																													
R-0																													
7															3			2			1			0					
Reserved															TX PDR		RX PDR		Reserved										
R-0															R/W-0		R/W-0		R/W-0										

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 29-29. SCI Pin I/O Control Register 6 (SCIPIO6) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX PDR	0 1	Transmit pin open drain enable. This bit enables open-drain capability in the output pin SCITX if the following conditions are met: - TX FUNC = 0 (SCITX pin is a general-purpose I/O.) - TX DIR = 1 (SCITX pin is a general-purpose output.) Open drain functionality is disabled; the output voltage is V_{OL} or lower if TXOUT = 0 and V_{OH} or higher if TXOUT = 1. Open drain functionality is enabled; the output voltage is V_{OL} or lower if TXOUT = 0 and high impedance if TXOUT = 1.
1	RX PDR	0 1	Receive pin open drain enable. This bit enables open-drain capability in the output pin SCIRX if the following conditions are met: - RX FUNC = 0 (SCIRX pin is a general-purpose I/O.) - RX DIR = 1 (SCIRX pin is a general-purpose output.) Open drain functionality is disabled; the output voltage is V_{OL} or lower if RXOUT = 0 and V_{OH} or higher if RXOUT = 1. Open drain functionality is enabled; the output voltage is V_{OL} or lower if RXOUT = 0 and high impedance if RXOUT = 1.
0	Reserved	0	Reads return 0. Writes have no effect.

29.7.20 SCI Pin I/O Control Register 7 (SCIPIO7)

Figure 29-29 and Table 29-30 illustrate this register.

Figure 29-29. SCI Pin I/O Control Register 7 (SCIPIO7) [offset = 58h]

31																													8
Reserved																													
R-0																													
7															3	2	1	0											
Reserved														TX PD		RX PD		Reserved											
R-0														R/W-n		R/W-n		R/W-n											

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset, Refer to the Terminal Functions in the device datasheet for default pin settings.

Table 29-30. SCI Pin I/O Control Register 7 (SCIPIO7) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX PD	0	Transmit pin pull control disable. This bit disables pull control capability on the input pin SCITX.
		0	The pull control on the SCITX pin is enabled.
		1	The pull control on the SCITX pin is disabled.
1	RX PD	0	Receive pin pull control disable. This bit disables pull control capability on the input pin SCIRX.
		0	Pull control on the SCIRX pin is enabled.
		1	Pull control on the SCIRX pin is disabled.
0	Reserved	0	Writes have no effect.

29.7.21 SCI Pin I/O Control Register 8 (SCIPIO8)

Figure 29-30 and Table 29-31 illustrate this register.

Figure 29-30. SCI Pin I/O Control Register 8 (SCIPIO8) [offset = 5Ch]

31																													8
Reserved																													
R-0																													
7															3	2	1	0											
Reserved														TX PSL		RX PSL		Reserved											
R-0														R/W-n		R/W-n		R/W-n											

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset, Refer to the Terminal Functions in the device datasheet for default pin settings.

Table 29-31. SCI Pin I/O Control Register 8 (SCIPIO8) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX PSL	0	TX pin pull select. This bit selects pull type in the input pin SCITX.
		0	The SCITX pin is a pull down.
		1	The SCITX pin is a pull up.
1	RX PSL	0	RX pin pull select. This bit selects pull type in the input pin SCIRX.
		0	The SCIRX pin is a pull down.
		1	The SCIRX pin is a pull up.
0	Reserved	0	Writes have no effect.

29.7.22 Input/Output Error Enable (IODFTCTRL) Register

Figure 29-31 and Table 29-32 illustrate this register. After the basic SCI module configuration, enable the required Error mode to be created followed by IODFT Key enable.

NOTE:

1. All the bits are used in IODFT mode only.
2. Each IODFT are expected to be checked individually.

Figure 29-31. Input/Output Error Enable Register (IODFTCTRL) [offset = 90h]

31	27			26	25	24
Reserved				FEN	PEN	BRKDTENA
R-0				R/W-0	R/WC-0	R/WC-0
23	21	20	19	18	16	
Reserved		PIN SAMPLE MASK		TX SHIFT		
R-0		R/W-0		R/W-0		
15	12		11	8		
Reserved			IODFTENA			
R-0			R/WP-0	R/WP-1	R/WP-0	R/WP-1
7				2	1	0
Reserved					LPB ENA	RXPENA
R-0					R/WP-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WC = Write in sci-compatible mode only; WP = Write in privilege mode only; -n = value after reset

Table 29-32. Input/Output Error Enable Register (IODFTCTRL) Field Descriptions

Bit	Field	Value	Description
31-27	Reserved	0	Reads return 0. Writes have no effect.
26	FEN	0	Frame error enable. This bit is used to create a frame error.
		1	No error is created.
		1	The stop bit received is ANDed with 0 and passed to the stop bit check circuitry.
25	PEN	0	Parity error enable. This bit is used to create a parity error.
		1	No parity error occurs.
		1	The parity bit received is toggled so that a parity error occurs.
24	BRKD TENA	0	Break detect error enable. This bit is used to create a BRKDT error.
		1	No error is created.
		1	The stop bit of the frame is ANDed with 0 and passed to the RSM so that a frame error occurs. Then the RX pin is forced to continuous low for 10 T _{BITS} so that a BRKDT error occurs.
32-21	Reserved	0	Reads return 0. Writes have no effect.
20-19	PIN SAMPLE MASK	0	Pin sample mask. These bits define the sample number at which the TX pin value that is being transmitted will be inverted to verify the receive pin samples majority detection circuitry.
		0	No mask is used.
		1h	Invert the TX Pin value at 7th.SCLK.
		2h	Invert the TX Pin value at 8th.SCLK.
		3h	Invert the TX Pin value at 9th.SCLK.

Table 29-32. Input/Output Error Enable Register (IODFTCTRL) Field Descriptions (continued)

Bit	Field	Value	Description
18-16	TX SHIFT		Transmit shift. These bits define the amount by which the value on TX pin is delayed so that the value on the RX pin is asynchronous. This feature is not applicable to the start bit.
		0	No delay occurs.
		1h	The value is delayed by 1 SCLK.
		2h	The value is delayed by 2 SCLK.
		3h	The value is delayed by 3 SCLK.
		4h	The value is delayed by 4 SCLK.
		5h	The value is delayed by 5 SCLK.
		6h	The value is delayed by 6 SCLK.
		7h	No delay occurs.
15-12	Reserved	0	Reads return 0. Writes have no effect.
11-8	IODFTENA		IODFT enable key. Write access permitted in Privilege mode only.
		Ah	IODFT is enabled.
		All Others	IODFT is disabled.
7-2	Reserved	0	Reads return 0. Writes have no effect.
1	LPBENA		Module loopback enable. Write access permitted in Privilege mode only. Note: In analog loopback mode the complete communication path through the I/Os can be tested, whereas in digital loopback mode the I/O buffers are excluded from this path.
		0	Digital loopback is enabled.
		1	Analog loopback is enabled in module I/O DFT mode when IODFTENA = 1010.
0	RXPENA		Module analog loopback through receive pin enable. Write access permitted in Privilege mode only. This bit defines whether the I/O buffers for the transmit or the receive pin are included in the communication path (in analog loopback mode).
		0	Analog loopback through the transmit pin is enabled.
		1	Analog loopback through the receive pin is enabled.

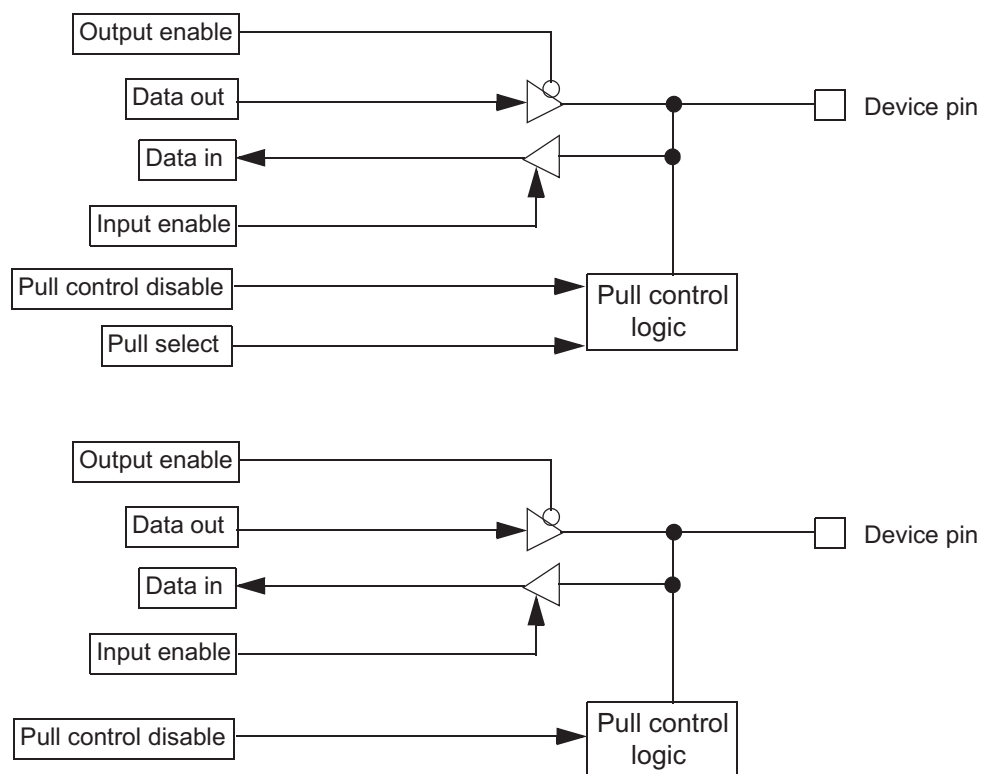
29.8 GPIO Functionality

The following sections apply to all device pins that can be configured as functional or general-purpose I/O pins.

29.8.1 GPIO Functionality

Figure 29-32 illustrates the GPIO functionality.

Figure 29-32. GPIO Functionality



29.8.2 Under Reset

The following apply if a device is under reset:

- Pull control. The reset pull control on the pins is enabled.
- Input buffer. The input buffer is enabled.
- Output buffer. The output buffer is disabled.

29.8.3 Out of Reset

The following apply if the device is out of reset:

- Pull control. The pull control is enabled by clearing the PD (pull control disable) bit in the SCIPIO7 register (Section 29.7.20). In this case, if the PSL (pull select) bit in the SCIPIO8 register (Section 29.7.21) is set, the pin will have a pull-up. If the PSL bit is cleared, the pin will have a pull-down. If the PD bit is set in the control register, there is no pull-up or pull-down on the pin.
- Input buffer. The input buffer is always enabled in functional mode.

NOTE: The pull-disable logic depends on the pin direction. It is independent of whether the device is in I/O or functional mode. If the pin is configured as output or transmit, then the pulls are disabled automatically. If the pin is configured as input or receive, the pulls are enabled or disabled depending on bit PD in the pull disable register SCIPIO7 (Section 29.7.20).

- Output buffer. A pin can be driven as an output pin if the TX DIR bit is set in the pin direction control register (SCIPIO1; Section 29.7.14) AND the open-drain feature is not enabled in the SCIPIO6 register (Section 29.7.19).

29.8.4 Open-Drain Feature Enabled on a Pin

The following apply if the open-drain feature is enabled on a pin:

- The output buffer is enabled, if a low signal is being driven on to the pin.
- The output buffer is disabled (the direction control signal DIR is internally forced low), if a high signal is being driven on to the pin.

NOTE: The open-drain feature is available only in I/O mode (SCIPIO0; Section 29.7.13).

29.8.5 Summary

The behavior of the input buffer, output buffer, and the pull control is summarized in Table 29-33.

Table 29-33. Input Buffer, Output Buffer, and Pull Control Behavior as GPIO Pins

Device under Reset?	Pin Direction (DIR) ⁽¹⁾⁽²⁾	Pull Disable (PULDIS) ⁽¹⁾⁽³⁾	Pull Select (PULSEL) ⁽¹⁾⁽⁴⁾	Pull Control	Output Buffer	Input Buffer
Yes	X	X	X	Enabled	Disabled	Enabled
No	0	0	0	Pull down	Disabled	Enabled
No	0	0	1	Pull up	Disabled	Enabled
No	0	1	0	Disabled	Disabled	Enabled
No	0	1	1	Disabled	Disabled	Enabled
No	1	X	X	Disabled	Enabled	Enabled

⁽¹⁾ X = Don't care

⁽²⁾ DIR = 0 for input, = 1 for output

⁽³⁾ PULDIS = 0 for enabling pull control
= 1 for disabling pull control

⁽⁴⁾ PULSEL = 0 for pull-down functionality
= 1 for pull-up functionality